REAL ANALYSIS II · SEMESTER 5 · MODULE 3

Answer key: The Riemann Integral

Formal solutions to all 40 questions of the Module 3 model paper (Bartle & Sherbert §7.1–§7.3).

BEFORE YOU READ AN ANSWER

Each answer starts with **Know first**: topic codes from File 1 (Prepare). If one feels new, read that topic first, try writing the skeleton yourself, and then read the solution.

Grey "Marks" notes show a typical split (a study guide, not an official scheme): statement · setup · key step · conclusion.

Section A · Short answers

3 marks each. Ceiling 48.

1 PYQ MTS-26 Q4 also MAT-22 Q3, MAT-21 Q5, MAT-22 MCQ 16

Definition of a partition and its norm; norms on [0,4] and [2,9].

KNOW FIRST

T01 M02

DEFINITION · PARTITION AND NORM (BARTLE §7.1, EQ. (1))

If $I := [a,b]$ is a closed bounded interval in $\mathbb{R}$, a **partition** of I is a finite, ordered set $P := (x_0, x_1, \dots, x_{n-1}, x_n)$ of points in I such that $a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b$. The **norm** (or mesh) of P is the length of the largest subinterval $I_i = [x_{i-1},x_i]$: $\lVert P\rVert := \max\{x_1 - x_0,\, x_2 - x_1,\, \dots,\, x_n - x_{n-1}\}$.

SKELETON

List $\Delta x_i = x_i - x_{i-1}$ for each partition; take $\lVert P\rVert = \max_i \Delta x_i$.

On [0,4]:

- $P_1 = (0,1,2,4)$: subinterval lengths are 1-0=1, 2-1=1, 4-2=2, so $\lVert P_1\rVert = \max\{1,1,2\} = 2$.
- $P_2 = (0,2,3,4)$: subinterval lengths are 2, 1, 1, so $\lVert P_2\rVert = \max\{2,1,1\} = 2$.
- $P_3 = (0,1,1.5,2,3.4,4)$: subinterval lengths are 1, 0.5, 0.5, 1.4, 0.6, so $\lVert P_3\rVert = \max\

{1,0.5,0.5,1.4,0.6\} = 1.4.

On [2,9] (MAT-22 MCQ 16): \lVert Q_1\rVert = \max\{2,2,2,1\} = 2; \lVert Q_2\rVert = \max\{4,3\} = 4; \lVert Q_3\rVert = \max\{3,2,2\} = 3. Thus Q_2 = (2,6,9) has the largest norm (MCQ option B).

ANSWER **\lVert P_1\rVert = 2,\ \lVert P_2\rVert = 2,\ \lVert P_3\rVert = 1.4; largest on [2,9] is Q_2 with \lVert Q_2\rVert = 4.**

Marks: definitions 1 · norms on [0,4] 1 · largest norm on [2,9] 1.

2 PYQ MTS-22 Q4

Tagged partition and Riemann sum; f(x) = x^2 on [0,4] with P = (0,1,2,4).

KNOW FIRST

T02 P02 M03

DEFINITION · TAGGED PARTITION AND RIEMANN SUM (BARTLE §7.1, EQ. (2))

If a point t_i \in I_i = [x_{i-1},x_i] is selected from each subinterval of a partition P for i=1,\dots,n, the points t_i are called **tags** and \dot P := \{([x_{i-1},x_i],t_i)\}_{i=1}^n is a **tagged partition** of [a,b]. For f : [a,b] \to \mathbb{R}, the **Riemann sum** of f corresponding to \dot P is S(f;\dot P) := \sum_{i=1}^n f(t_i)(x_i - x_{i-1}).

SKELETON

Write the subintervals [0,1], [1,2], [2,4] with widths \Delta x_1 = 1, \Delta x_2 = 1, \Delta x_3 = 2, evaluate f(t_i) = t_i^2, and sum f(t_1)\Delta x_1 + f(t_2)\Delta x_2 + f(t_3)\Delta x_3.

Left-endpoint tags t_1 = 0, t_2 = 1, t_3 = 2: S(f;\dot P_{\text{left}}) = 0^2(1 - 0) + 1^2(2 - 1) + 2^2(4 - 2) = 0 + 1 + 8 = 9.

Right-endpoint tags t_1 = 1, t_2 = 2, t_3 = 4: S(f;\dot P_{\text{right}}) = 1^2(1 - 0) + 2^2(2 - 1) + 4^2(4 - 2) = 1 + 4 + 32 = 37.

ANSWER **S(f;\dot P_{\text{left}}) = 9 and S(f;\dot P_{\text{right}}) = 37.**

Marks: definition 1 · left-endpoint sum 1 · right-endpoint sum 1.

3 PYQ MAT-21 MCQ 9 · adapted

Definition of f \in \mathcal{R}[a,b]; why \int_a^b f is not an ordinary function limit.

KNOW FIRST

T03 M02 M03

DEFINITION 7.1.1 · RIEMANN INTEGRABLE (BARTLE P. 201)

A function f : [a,b] \to \mathbb{R} is said to be **Riemann integrable** on [a,b] if there exists a number L \in \mathbb{R} such that for every \varepsilon > 0 there exists \delta_\varepsilon > 0 such that if \dot P is any tagged partition of [a,b] with \lVert P\rVert < \delta_\varepsilon, then \lvert S(f;\dot P) - L\rvert < \varepsilon. The set of all Riemann integrable functions on [a,b] is denoted by \mathcal{R}[a,b], and we write L = \int_a^b f or \int_a^b f(x)\,dx.

Why this is not an ordinary function limit as \lVert P\rVert \to 0 (Remark 7.1.1): For a given positive number h = \lVert P\rVert, there are infinitely many partitions P with norm h and infinitely many choices of tags t_i \in I_i for each such P. Because many different Riemann sums S(f;\dot P) correspond to the same number \lVert P\rVert, S(f;\dot P) is not a single-valued function of \lVert P\rVert. (In MAT-21 MCQ 9, the function satisfying Definition 7.1.1 is called **Riemann integrable**, option A.)

Marks: Definition 7.1.1 word for word 1.5 · notation \mathcal{R}[a,b] and \int_a^b f 0.5 · non-function limit

explanation 1.

4

Uniqueness of the Riemann integral (Theorem 7.1.2).

KNOW FIRST

T04 P05 Triangle inequality and the "arbitrary ε" rule M08

SKELETON

Assume L' and L'' both satisfy Definition 7.1.1. Given \varepsilon > 0, take \delta'_{\varepsilon/2} and \delta''_{\varepsilon/2}, put \delta_\varepsilon = \min\{\delta'_{\varepsilon/2},\delta''_{\varepsilon/2}\}, and pick one \dot P with \lVert P\rVert < \delta_\varepsilon to get \lvert L' - L''\rvert < \varepsilon.

PROOF OF THEOREM 7.1.2 (BARTLE P. 201)

1. Suppose L' and L'' both satisfy Definition 7.1.1 for f \in \mathcal{R}[a,b], and let \varepsilon > 0 be arbitrary.
   We will show \lvert L' - L''\rvert < \varepsilon for every \varepsilon > 0 (P05).

2. By Definition 7.1.1 with tolerance \varepsilon/2 > 0, there exists \delta'_{\varepsilon/2} > 0 such that \lVert P_1\rVert < \delta'_{\varepsilon/2} \Rightarrow \lvert S(f;\dot P_1) - L'\rvert < \varepsilon/2, and there exists \delta''_{\varepsilon/2} > 0 such that \lVert P_2\rVert < \delta''_{\varepsilon/2} \Rightarrow \lvert S(f;\dot P_2) - L''\rvert < \varepsilon/2.
   Definition 7.1.1 applied to L' and L''.

3 Put $\delta_\varepsilon := \min\{\delta'_{\varepsilon/2}, \delta''_{\varepsilon/2}\} > 0$, and choose a tagged partition $\dot P$ of $[a,b]$ with $\lVert P\rVert < \delta_\varepsilon$. Then both inequalities of Step 2 hold for $\dot P$, so by the Triangle Inequality, $\lvert L' - L''\rvert \le \lvert L' - S(f;\dot P)\rvert + \lvert S(f;\dot P) - L''\rvert < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $L' = L''$. ∎

Common δ_ε argument (M08), Triangle Inequality, and arbitrary ε rule (P05).

Marks: setup with $\varepsilon/2$ 1 · $\delta_\varepsilon = \min\{\delta'_{\varepsilon/2},\delta''_{\varepsilon/2}\}$ and triangle inequality 1.5 · conclusion 0.5.

5 PYQ MTS-25 Q4

Constant function $f(x) = k$ is in $\mathcal{R}[a,b]$ with $\int_a^b k = k(b-a)$.

KNOW FIRST

T05 T03 M01

SKELETON

Compute $S(f;\dot P) = k(b-a)$ for every tagged partition $\dot P$; given $\varepsilon > 0$, take $\delta_\varepsilon = 1$.

PROOF (BARTLE EXAMPLE 7.1.4(A), P. 202)

1. Let f(x) := k for all x \in [a,b]. If \dot P = \{([x_{i-1},x_i],t_i)\}_{i=1}^n is any tagged partition of [a,b], then S(f;\dot P) = \sum_{i=1}^n f(t_i)(x_i - x_{i-1}) = \sum_{i=1}^n k(x_i - x_{i-1}) = k\sum_{i=1}^n (x_i - x_{i-1}) = k(x_n - x_0) = k(b - a).
   Factor out k (P02) and telescope \sum_{i=1}^n (x_i - x_{i-1}) = b - a (M01).

2. Hence for any \varepsilon > 0, we can choose \delta_\varepsilon := 1 > 0 so that if \dot P has \lVert P\rVert < \delta_\varepsilon, then \lvert S(f;\dot P) - k(b-a)\rvert = \lvert k(b-a) - k(b-a)\rvert = 0 < \varepsilon. Since \varepsilon > 0 is arbitrary, f \in \mathcal{R}[a,b] and \int_a^b k = k(b-a). ∎
   Definition 7.1.1 satisfied with L = k(b-a) and \delta_\varepsilon = 1.

ANSWER $\displaystyle\int_a^b k = k(b - a).$

Marks: telescoping Riemann sum S(f;\dot P) = k(b-a) 1.5 · explicit \delta_\varepsilon = 1 and \lvert S - k(b-a)\rvert = 0 < \varepsilon 1 · conclusion 0.5.

6

PYQ MAT-22 MCQ 18 · adapted

also MAT-21 MCQ 13, MTS-22 MCQ 7 · adapted

Changing g \in \mathcal{R}[a,b] at finitely many points (Theorem 7.1.3); \int_0^2 h; finite vs countable exception sets.

KNOW FIRST

T06 M03 M04

SKELETON

For one exceptional point c, bound \lvert S(f;\dot P) - S(g;\dot P)\rvert \le 2(\lvert f(c)\rvert + \lvert g(c)\rvert)\lVert P\rVert < \varepsilon/2 when \lVert P\rVert < \delta_1, and combine with \lvert S(g;\dot P) - L\rvert < \varepsilon/2.

PROOF OF THEOREM 7.1.3 (BARTLE PP. 201–202)

1. It suffices to prove the case of one exceptional point $c \in [a,b]$ (finitely many points follow by induction). Let $L := \int_a^b g$ and suppose $f(x) = g(x)$ for all $x \ne c$.
   Induction reduces m exceptional points to 1 point.

2. For any tagged partition $\dot P$, the sums $S(f;\dot P)$ and $S(g;\dot P)$ agree except on at most two subintervals (when c is a partition point tag of two adjacent intervals). Thus $\lvert S(f;\dot P) - S(g;\dot P)\rvert \le 2\big(\lvert f(c)\rvert + \lvert g(c)\rvert\big)\lVert P\rVert$.
   At most 2 terms differ (M03), each of width $\le \lVert P\rVert$ (M04).

3. Given \varepsilon > 0, choose \delta_1 > 0 with \delta_1 < \varepsilon/\big(4(\lvert f(c)\rvert + \lvert g(c)\rvert + 1)\big) and choose \delta_2 > 0 so that \lVert P\rVert < \delta_2 \Rightarrow \lvert S(g;\dot P) - L\rvert < \varepsilon/2. Let \delta := \min\{\delta_1,\delta_2\} > 0. For \lVert P\rVert < \delta, \lvert S(f;\dot P) - L\rvert \le \lvert S(f;\dot P) - S(g;\dot P)\rvert + \lvert S(g;\dot P) - L\rvert < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, so f \in \mathcal{R}[a,b] and \int_a^b f = L = \int_a^b g. ∎

Triangle Inequality and Definition 7.1.1.

Applications & MCQ resolutions:

- For h(x) = 2 on [0,1), h(1) = 3, h(x) = 1 on (1,2]: h differs from the step function g = 2\varphi_{[0,1]} + 1\varphi_{(1,2]} only at c = 1, so □\int_0^2 h = \int_0^2 g = 2(1) + 1(1) = 3.
- If f(x) = 0 except at finitely many points c_1,\dots,c_n (MAT-22 MCQ 18, MAT-21 MCQ 13): comparing with g \equiv 0 gives **both** f \in \mathcal{R}[a,b] (option A) **and** \int_a^b f = 0 (option B). (Note: in MAT-22 MCQ 18 both A and B are true; in MAT-21 MCQ 13 option B is the unique correct option.)
- **Ambiguous past question note (MTS-22 MCQ 7):** When f(x) = g(x) off the **countably infinite** set \{1/n : n \in \mathbb{N}\}, knowing f \in \mathcal{R}[0,1] does **not** guarantee g \in

\mathcal{R}[0,1] in general (e.g. f = 0 and g(1/n) = n is unbounded, so g \notin \mathcal{R}[0,1]); if both f, g \in \mathcal{R}[0,1] then \int_0^1 f = \int_0^1 g since f g = 0 outside a null set. The intended exam key for MTS-22 MCQ 7 treats the unquantified statement (C) as the false option, or assumes boundedness.

ANSWER **\displaystyle\int_0^2 h = 3; if f=0 off a finite set, f\in\mathcal{R}[a,b] and \int_a^b f = 0.**

Marks: proof of 7.1.3 1.5 · \int_0^2 h = 3 0.5 · finite vs countable discussion 1.

7

PYQ MAT-22 MCQ 15 · adapted **also MAT-21 MCQ 11, 17, MTS-22 MCQ 8, 9 · adapted**

Boundedness Theorem 7.1.6; H(1/k) = k not integrable; converses.

KNOW FIRST

T08 P05

THEOREM 7.1.6 · BOUNDEDNESS THEOREM (BARTLE P. 205)

If f \in \mathcal{R}[a,b], then f is bounded on [a,b].

SKELETON

Suppose f \in \mathcal{R}[a,b] is unbounded. Fix \delta > 0 for \varepsilon = 1 so \lvert S(f;\dot P)\rvert < \lvert L\rvert + 1. Fix a partition Q with \lVert Q\rVert < \delta; f is unbounded on some I_k; pick t_k \in I_k huge to contradict \lvert S(f;\dot Q)\rvert < \lvert L\rvert + 1.

PROOF OF THEOREM 7.1.6 (BARTLE PP. 205–206)

1. Suppose f \in \mathcal{R}[a,b] with integral L, and assume for contradiction that f is unbounded on [a,b]. Taking \varepsilon = 1 in Definition 7.1.1 gives \delta > 0 such that \lVert P\rVert < \delta \Rightarrow \lvert S(f;\dot P) - L\rvert < 1, whence \lvert S(f;\dot P)\rvert < \lvert L\rvert + 1.
   Definition 7.1.1 with \varepsilon = 1 and Triangle Inequality.

2. Let Q = \{[x_{i-1},x_i]\}_{i=1}^n be a partition of [a,b] with \lVert Q\rVert < \delta. Since f is unbounded on [a,b], it must be unbounded on at least one subinterval [x_{k-1},x_k].
   If f were bounded on each of the n pieces by M_i, it would be bounded on [a,b] by \max_i M_i.

3. Tag Q by setting t_i := x_i for i \ne k, and choose t_k \in [x_{k-1},x_k] so that \lvert f(t_k)(x_k - x_{k-1})\rvert > \lvert L\rvert + 1 + \left\lvert\sum_{i\ne k} f(t_i)(x_i x_{i-1})\right\rvert. By the Reverse Triangle Inequality, \lvert S(f;\dot Q)\rvert \ge \lvert f(t_k)(x_k - x_{k-1})\rvert \big\lvert\sum_{i\ne k} f(t_i)(x_i - x_{i-1})\big\rvert > \lvert L\rvert + 1, contradicting Step 1. Thus f is bounded. ∎
   Reverse Triangle Inequality (P05).

Applications & Converses:

- For H(1/k) = k (k \in \mathbb{N}), H(x) = 0 elsewhere on [0,1]: since H(1/k) = k \to \infty, H is unbounded on [0,1], so by Theorem 7.1.6, H \notin \mathcal{R}[0,1] (MTS-22 MCQ 9 option A; MAT-21 MCQ 17 option B; MTS-22 MCQ 8 option D).
- f \in \mathcal{R}[a,b] does **not** imply f is continuous or monotone (Thomae's function C8 is integrable but neither continuous nor monotone; MAT-22 MCQ 15 and MAT-21 MCQ 11 option A).
- Not every bounded function is integrable: Dirichlet's function C7 is bounded (0 \le f \le 1) but f \notin \mathcal{R}[0,1].

Marks: statement & proof of 7.1.6 2 · H \notin \mathcal{R}[0,1] and converses 1.

8 PYQ MAT-21 MCQ 10 · adapted

Elementary step function \varphi_J (Lemma 7.2.4) and \int_0^4 g.

KNOW FIRST

T13 T06 M10

SKELETON

For J = [c,d] and \lVert P\rVert < \delta, trap S(\varphi_{[c,d]};\dot P) between (d-c)-2\delta and (d-c)+2\delta and pick \delta_\varepsilon = \varepsilon/4; extend to (c,d), [c,d), (c,d] via Theorem 7.1.3.

PROOF OF LEMMA 7.2.4 (BARTLE P. 210)

1. First let J = [c,d] \subseteq [a,b] (c < d). For any tagged partition \dot P with \lVert P\rVert < \delta, the union U_1 of subintervals in \dot P with tag t_i \in [c,d] satisfies [c+\delta, d-\delta] \subseteq U_1 \subseteq [c-\delta, d+\delta] (M10), so (d - c) - 2\delta \le S(\varphi_{[c,d]};\dot P) \le (d - c) + 2\delta.
   Tag-inclusion bounds (M10 / Exercise 7.1.13).

2. Given \varepsilon > 0, take \delta_\varepsilon := \varepsilon/4 > 0; then \lVert P\rVert < \delta_\varepsilon \Rightarrow \lvert S(\varphi_{[c,d]};\dot P) - (d - c)\rvert \le 2(\varepsilon/4) = \varepsilon/2 < \varepsilon. Thus \varphi_{[c,d]} \in \mathcal{R}[a,b] and \int_a^b \varphi_{[c,d]} = d - c.
   Definition 7.1.1 with \delta_\varepsilon = \varepsilon/4.

3. The other three intervals (c,d), [c,d), (c,d] have indicator functions differing from \varphi_{[c,d]} at at most the two endpoints c, d. By Theorem 7.1.3, each \varphi_J \in \mathcal{R}[a,b] with \int_a^b \varphi_J = d - c. ∎
   Theorem 7.1.3 (T06).

Evaluation of \int_0^4 g (MAT-21 MCQ 10): Since g(x) = 3 on [0,2] and g(x) = 2 on (2,4], we write g = 3\varphi_{[0,2]} + 2\varphi_{(2,4]}. By Lemma 7.2.4

and Theorem 7.1.5, \int_0^4 g = 3(2 - 0) + 2(4 - 2) = 6 + 4 = 10 (MCQ option C).

ANSWER **\displaystyle\int_a^b \varphi_J = d - c and \displaystyle\int_0^4 g = 10.**

Marks: proof for [c,d] 1.5 · extension by 7.1.3 0.5 · \int_0^4 g = 10 1.

9

Oriented integrals (Definition 7.2.12, Theorem 7.2.13) and check for f(x) = x, \alpha = 3, \beta = 1, \gamma = 0.

KNOW FIRST

T19 T17

DEFINITION 7.2.12 & THEOREM 7.2.13 (BARTLE P. 214)

If f \in \mathcal{R}[a,b] and \alpha, \beta \in [a,b] with \alpha < \beta, we define \int_\beta^\alpha f := -\int_\alpha^\beta f \qquad \text{and} \qquad \int_\alpha^\alpha f := 0. **Theorem 7.2.13:** For any numbers \alpha, \beta, \gamma \in [a,b], \displaystyle\int_\alpha^\beta f = \int_\alpha^\gamma f + \int_\gamma^\beta f.

Verification for f(x) = x with \alpha = 3, \beta = 1, \gamma = 0: Using \int_u^v x\,dx = \tfrac12(v^2 - u^2) for u < v:

- \displaystyle\int_3^1 x\,dx = -\int_1^3 x\,dx = -\tfrac12(3^2 - 1^2) = -4.

- \displaystyle\int_3^0 x\,dx + \int_0^1 x\,dx = -\int_0^3 x\,dx + \int_0^1 x\,dx = -\tfrac12(9 - 0) + \tfrac12(1 - 0) = -\frac92 + \frac12 = -4.

ANSWER **Both sides equal -4.**

Marks: Definition 7.2.12 1 · Theorem 7.2.13 statement 0.5 · verification 1.5.

10

Why FTC 1 fails for H(x) = 2\sqrt x on [0,b] and K(x) = x^2\cos(1/x^2) on [0,1].

KNOW FIRST
T21 T20 T08

Theorem 7.3.1 (FTC 1) requires three hypotheses: (a) F continuous on [a,b]; (b) F'(x) = f(x) on [a,b]\setminus E for a finite set E; and (c) f \in \mathcal{R}[a,b]. In both examples, **hypothesis (c) fails** because the derivative is unbounded (Theorem 7.1.6):

- **For H(x) = 2\sqrt x on [0,b] (Example 7.3.2(d)):** H is continuous on [0,b] and H'(x) = \frac{1}{\sqrt x} on (0,b) (E = \{0\}). However, h(x) = \frac{1}{\sqrt x} is **unbounded** on (0,b) as x \to 0^+, so by Theorem 7.1.6, h \notin \mathcal{R}[0,b] regardless of how h(0) is defined.
- **For K(x) = x^2\cos(1/x^2) (x \in (0,1]), K(0) = 0 on [0,1] (Example 7.3.2(e), C19):** For x > 0, the Product and Chain Rules give K'(x) = 2x\cos(1/x^2) + \frac{2}{x}\sin(1/x^2), and at x = 0, K'(0) = \lim_{x\to 0}\frac{x^2\cos(1/x^2)}{x} =

0. Thus K is continuous and differentiable at **every** point of [0,1] (E = \varnothing). Yet along x_m = (\frac{\pi}{2} + 2m\pi)^{-1/2} \to 0^+, \sin(1/x_m^2) = 1 and K'(x_m) = \frac{2}{x_m} \to \infty, so K' is unbounded on [0,1] and K' \notin \mathcal{R}[0,1].

Marks: identify hypothesis (c) f\in\mathcal{R} via 7.1.6 1 · H'(x)=1/\sqrt x unbounded 1 · K'(x) formula, K'(0)=0, and unboundedness 1.

11 PYQ MAT-21 Q6

Definition of the indefinite integral F(z) = \int_a^z f; proof that F is Lipschitz and continuous (Theorem 7.3.4).

KNOW FIRST

T22 T17 T07

DEFINITION 7.3.3 · INDEFINITE INTEGRAL (BARTLE P. 218)

If f \in \mathcal{R}[a,b], the function defined by F(z) := \int_a^z f \quad \text{for } z \in [a,b] is called the **indefinite integral** of f with **basepoint** a.

PROOF OF THEOREM 7.3.4 (BARTLE P. 218)

1. Let w, z \in [a,b] with w \le z. By the Additivity Theorem 7.2.9, F(z) = \int_a^z f = \int_a^w f + \int_w^z f = F(w) + \int_w^z f, so F(z) - F(w) = \int_w^z f.
   Additivity Theorem 7.2.9 (T17).

2. Since \lvert f(x)\rvert \le M on [a,b], we have -M \le f(x) \le M on [w,z]. By Theorem 7.1.5(c) and Example 7.1.4(a), -M(z - w) \le \int_w^z f \le M(z - w), whence \lvert F(z) - F(w)\rvert = \left\lvert\int_w^z f\right\rvert \le M\lvert z - w\rvert for all z, w \in [a,b].
   Monotonicity of the integral (7.1.5c) and constant integral (7.1.4a).

3. Thus F is Lipschitz on [a,b] with constant M, and hence uniformly continuous and continuous on [a,b] (given \varepsilon > 0, take \delta := \varepsilon/(M+1)). ∎
   Lipschitz \Rightarrow continuous (P09).

Marks: Definition 7.3.3 1 · Additivity step 1 · Lipschitz bound and continuity 1.

12

Indefinite integrals of \operatorname{sgn} x on [-1,1] and Thomae's h on [0,1] (Examples 7.3.7(a), (b)).

KNOW FIRST

T24 T23 T09

1. For f(x) = \operatorname{sgn} x on [-1,1] with basepoint -1 (Example 7.3.7(a), C20):

- If -1 \le x < 0, then f(t) = -1 on [-1,x), so F(x) = \int_{-1}^x (-1)\,dt = -(x + 1) = \lvert x\rvert - 1.
- If 0 \le x \le 1, Additivity gives F(x) = \int_{-1}^0 (-1)\,dt + \int_0^x 1\,dt = -1 + x = \lvert x\rvert - 1.

Thus F(x) = \lvert x\rvert - 1 for all x \in [-1,1]. Since F'_-(0) = -1 \ne +1 = F'_+(0), F'(0) does not exist, so F is **not** an antiderivative of f on [-1,1].

2. For Thomae's function h on [0,1] (Example 7.3.7(b), C8): Since \int_0^1 h = 0 and h(x) \ge 0, we have 0 \le H(x) = \int_0^x h \le \int_0^1 h = 0, so H(x) = 0 for all x \in [0,1]. Although H'(x) = 0 exists everywhere on [0,1], at every rational x = m/n \in \mathbb{Q}\cap[0,1] we have H'(x) = 0 \ne 1/n = h(x), so H is **not** an antiderivative of h on [0,1].

ANSWER **F(x) = \lvert x\rvert - 1 (F'(0) DNE); H(x) = 0 (H'(x) \ne h(x) on \mathbb{Q}\cap[0,1]). Neither is an antiderivative.**

Marks: F(x) = \lvert x\rvert - 1 on [-1,1] 1 · F'(0) DNE 1 · Thomae H \equiv 0 and H' \ne h on rationals 1.

13 PYQ MTS-25 Q7 also MTS-26 Q7, MTS-22 MCQ 10

Definition of a null set; proof that \mathbb{Q}\cap[0,1] is null; properties of null sets.

KNOW FIRST

T26 P03 P13 M07

DEFINITION 7.3.10(A) · NULL SET (BARTLE P. 220)

A set $Z \subseteq \mathbb{R}$ is said to be a **null set** if for every $\varepsilon > 0$ there exists a countable collection $\{(a_k,b_k)\}_{k=1}^\infty$ of open intervals such that $Z \subseteq \bigcup_{k=1}^\infty (a_k,b_k) \qquad \text{and} \qquad \sum_{k=1}^\infty (b_k a_k) \le \varepsilon$.

PROOF THAT \MATHBB{Q}_1 = \MATHBB{Q}\CAP[0,1] IS A NULL SET (EXAMPLE 7.3.11, P. 221)

1. Since \mathbb{Q}_1 = \mathbb{Q}\cap[0,1] is countable, enumerate its elements as \mathbb{Q}_1 = \{r_1, r_2, \dots, r_k, \dots\}.
   Countability of \mathbb{Q}\cap[0,1] (P13).

2. Given \varepsilon > 0, for each k \in \mathbb{N} define the open interval J_k := \left(r_k - \dfrac{\varepsilon}{2^{k+1}},\, r_k + \dfrac{\varepsilon}{2^{k+1}}\right), which contains r_k and has length \dfrac{\varepsilon}{2^k}.
   Geometric halving trick (M07).

3. Then \mathbb{Q}_1 \subseteq \bigcup_{k=1}^\infty J_k and the total length is \sum_{k=1}^\infty \dfrac{\varepsilon}{2^k} = \varepsilon\sum_{k=1}^\infty \dfrac{1}{2^k} = \varepsilon. Since \varepsilon > 0 is arbitrary, \mathbb{Q}\cap[0,1] is a null set. ∎
   Geometric series \sum_{k=1}^\infty 1/2^k = 1 (P03).

Properties (MTS-22 MCQ 10): Any subset of a null set is null (A true); the union of two null sets is null (B true); every countable set is null by the same proof (C true). However, **not every null set is**

countable (D is the **false** statement): the Cantor set is an uncountable null set.

Marks: Definition 7.3.10(a) 1 · proof for \mathbb{Q}\cap[0,1] 1.5 · properties & Cantor set remark 0.5.

14 PYQ MAT-22 MCQ 27, 28, 29 · adapted | also MAT-21 MCQ 19, 20 · adapted

FTC 1 evaluations: \int_a^b f = F(b)-F(a), \int_{-2}^3 g = 5/2, and \int_a^b \lvert x\rvert\,dx = B(b)-B(a).

KNOW FIRST

T20 T21

(a) MAT-22 MCQ 27 / MAT-21 MCQ 19: If f is continuous on [a,b] (hence f \in \mathcal{R}[a,b] by 7.2.7) and F is any antiderivative of f on [a,b] (F is continuous by 6.1.2 and F' = f with E = \varnothing), then by Theorem 7.3.1, \int_a^b f(x)\,dx = F(b) - F(a) (MCQ option B).

(b) Bartle Ex 7.3.3 (MAT-22 MCQ 28 / MAT-21 MCQ 20): On [-2,3], G(x) = \tfrac12\lvert x^2 - 1\rvert is continuous. For \lvert x\rvert > 1, G(x) = \tfrac12(x^2-1) has G'(x) = x = g(x); for \lvert x\rvert < 1, G(x) = \tfrac12(1-x^2) has G'(x) = -x = g(x). Thus G'(x) = g(x) off the finite set E = \{-1, 1\}, and g \in \mathcal{R}[-2,3] (piecewise continuous). By FTC 1, \int_{-2}^3 g(x)\,dx = G(3) - G(-2) = \frac12\lvert 9 - 1\rvert - \frac12\lvert 4 - 1\rvert = 4 - \frac32 = \frac52 \quad \text{(MCQ option B)}.

(c) Bartle Ex 7.3.4 (MAT-22 MCQ 29): For B(x) = -\tfrac12 x^2 (x < 0), \tfrac12 x^2 (x \ge 0), we have B'(x) = -x = \lvert x\rvert for x < 0, B'(x) = x = \lvert

x\rvert for x > 0, and □B'_-(0) = 0 = B'_+(0) = \lvert 0\rvert. Thus B is continuous and B'(x) = \lvert x\rvert on all of [a,b] with E = \varnothing, and \lvert x\rvert \in \mathcal{R}[a,b] (continuous). By FTC 1, \int_a^b \lvert x\rvert\,dx = B(b) - B(a) (MCQ option B).

ANSWER **(a) F(b)-F(a); (b) \displaystyle\int_{-2}^3 g = \frac52 (E=\{-1,1\}); (c) \displaystyle\int_a^b \lvert x\rvert\,dx = B(b)-B(a) (E=\varnothing).**

Marks: part (a) 0.5 · part (b) with E=\{-1,1\} and value 5/2 1.5 · part (c) with E=\varnothing 1.

15 PYQ MAT-22 Q35

Evaluate \int_0^1 x^2(1 - \sqrt x)\,dx using the Fundamental Theorem of Calculus.

KNOW FIRST

T20 T21 T15

SKELETON

Write f(x) = x^2 - x^{5/2} and F(x) = \tfrac13 x^3 - \tfrac27 x^{7/2} on [0,1]. Check hypotheses (a), (b), (c) of Theorem 7.3.1 with E = \varnothing, then compute F(1) - F(0).

SOLUTION

1. On [0,1], expand the integrand as f(x) := x^2(1 - \sqrt x) = x^2 - x^{5/2}. Since f is continuous on the closed bounded interval [0,1], Theorem 7.2.7 implies that f \in \mathcal{R}[0,1] (hypothesis (c) of Theorem 7.3.1).
   Continuous \Rightarrow integrable (Theorem 7.2.7, T15).

2. Define F : [0,1] \to \mathbb{R} by F(x) := \frac{x^3}{3} - \frac{2}{7}x^{7/2}. Then F is continuous on [0,1] (hypothesis (a)), and F'(x) = x^2 - x^{5/2} = f(x) for all x \in [0,1] (hypothesis (b) with E = \varnothing).
   Power rule for derivatives on [0,1].

3. By the First Form of the Fundamental Theorem of Calculus (Theorem 7.3.1), \int_0^1 x^2(1 - \sqrt x)\,dx = F(1) - F(0) = \left(\frac{1}{3} - \frac{2}{7}\right) - 0 = \frac{7 - 6}{21} = \frac{1}{21}. \quad \blacksquare
   Theorem 7.3.1 (T20).

ANSWER **\displaystyle\int_0^1 x^2(1 - \sqrt x)\,dx = \frac{1}{21}.**

Marks: f \in \mathcal{R}[0,1] by continuity 1 · antiderivative F(x) = \tfrac13 x^3 \tfrac27 x^{7/2} 1 · F(1)-F(0) = 1/21 1.

16 PYQ MAT-22 MCQ 19, 26 · adapted

Trigonometric symmetry \int_0^{\pi/2} f(\cos x)\,dx and translation \int_{a+c}^{b+c} f(y-c)\,dy = \int_a^b f.

KNOW FIRST

T25 T19

(a) MAT-22 MCQ 26: Let \varphi(x) := \tfrac{\pi}{2} - x on [0,\pi/2]. Since \varphi'(x) = -1 is continuous and \cos x = \sin(\tfrac{\pi}{2} - x), Theorem 7.3.8 and Definition 7.2.12 give \int_0^{\pi/2} f(\cos x)\,dx = -\int_0^{\pi/2} f(\sin(\varphi(x)))\,\varphi'(x)\,dx = -\int_{\pi/2}^0 f(\sin u)\,du = \int_0^{\pi/2} f(\sin u)\,du. Next, on [\pi/2,\pi], substituting u = \psi(x) := \pi - x (\psi' = -1, \sin x = \sin(\pi - x)) gives \int_{\pi/2}^\pi f(\sin x)\,dx = \int_0^{\pi/2} f(\sin u)\,du. By Additivity 7.2.9, \int_0^\pi f(\sin x)\,dx = 2\int_0^{\pi/2} f(\sin x)\,dx, so \int_0^{\pi/2} f(\cos x)\,dx = \tfrac12\int_0^\pi f(\sin x)\,dx. Thus options (A) and (B) are equal to \int_0^{\pi/2} f(\cos x)\,dx, while **(C) \int_0^\pi f(\sin x)\,dx is not equal to it** (twice as large whenever the integral is non-zero).

(b) Bartle Ex 7.1.15 / MAT-22 MCQ 19: For g(y) = f(y - c) on [a+c,b+c], any tagged partition \dot P_c = \{([y_{i-1},y_i],s_i)\} of [a+c,b+c] shifts by -c to a tagged partition \dot P = \{([y_{i-1}-c,y_i-c],s_i-c)\} of [a,b] with the same norm and S(g;\dot P_c) = S(f;\dot P). Thus g \in \mathcal{R}[a+c,b+c] and \int_{a+c}^{b+c} g = \int_a^b f (MCQ option B; when f is continuous this also follows from

Theorem 7.3.8 with \varphi(y) = y - c, \varphi'(y) = 1).

ANSWER **(a) Not equal to (C) \int_0^\pi f(\sin x)\,dx; (b) \displaystyle\int_{a+c}^{b+c} g = \int_a^b f (option B).**

Marks: \int_0^{\pi/2} f(\cos x)\,dx = \int_0^{\pi/2} f(\sin x)\,dx 1 · \tfrac12\int_0^\pi f(\sin x)\,dx 1 · translate formula 1.

17 PYQ MAT-22 MCQ 30 · adapted

If f : [0,1] \to \mathbb{R} is continuous and \int_0^x f = \int_x^1 f for all x \in [0,1], then f \equiv 0.

KNOW FIRST

T23 T22 T17 T15

SKELETON

Set F(x) = \int_0^x f; use Additivity to write \int_x^1 f = F(1) - F(x), deduce F(x) = \tfrac12 F(1) is constant, and differentiate via Theorem 7.3.6.

PROOF (BARTLE EXERCISE 7.3.16)

1. Since f is continuous on [0,1], Theorem 7.2.7 implies that f \in \mathcal{R}[0,1]. Define the indefinite integral F(x) := \int_0^x f for x \in [0,1].
   Theorem 7.2.7 (T15) proves (B) f \in \mathcal{R}[0,1].

2. By the Additivity Theorem 7.2.9, F(1) = \int_0^1 f = \int_0^x f + \int_x^1 f. Since \int_0^x f = \int_x^1 f by hypothesis, we have 2F(x) = F(1), so F(x) = \tfrac12 F(1) is constant on [0,1] (in fact, setting x = 0 gives F(1) = 2F(0) = 0, so F(x) = 0 for all x \in [0,1]).
   Additivity Theorem 7.2.9 (T17).

3. Since f is continuous on [0,1], Theorem 7.3.6 (FTC 2) gives f(x) = F'(x) = 0 for all x \in [0,1]. ∎
   Theorem 7.3.6 (T23). (In MAT-22 MCQ 30, both (A) f(x)=0\ \forall x and (B) f\in\mathcal{R}[0,1] hold, so the answer is (C) "All the above".)

Marks: f \in \mathcal{R}[0,1] 0.5 · 2F(x) = F(1) \Rightarrow F \equiv 0 via Additivity 1.5 · f = F' = 0 via FTC 2 1.

18 PYQ MAT-22 MCQ 23 · adapted

If f is continuous on [a,b], f \ge 0, and \int_a^b f = 0, then f(x) = 0 for all x \in [a,b].

KNOW FIRST

T15 T13 T07

SKELETON

Suppose f(c) > 0 for some c \in [a,b]. Use continuity at c with \varepsilon = \tfrac12 f(c) to find a closed subinterval J \subseteq [a,b] of length d-c > 0 where f \ge \tfrac12 f(c), getting \int_a^b f \ge \tfrac12 f(c)(d-c) > 0.

PROOF (BARTLE EXERCISES 7.2.8–7.2.9)

1. Suppose for contradiction that there is some c \in [a,b] with f(c) > 0. Since f is continuous at c, taking \varepsilon := \tfrac12 f(c) > 0 gives \delta > 0 such that if x \in [a,b] and \lvert x - c\rvert < \delta, then f(x) > f(c) - \tfrac12 f(c) = \tfrac12 f(c).
   Definition of continuity at c with \varepsilon = \tfrac12 f(c).

2. Choose a non-degenerate closed subinterval J = [c_1, d_1] \subseteq [a,b] \cap (c-\delta, c+\delta) with c_1 < d_1. Since f(x) \ge 0 on [a,b] and f(x) > \tfrac12 f(c) on J, we have \tfrac12 f(c)\varphi_J(x) \le f(x) for all x \in [a,b].
   Lower bound by an elementary step function (M06).

3. By Lemma 7.2.4 and Theorem 7.1.5(c), \int_a^b f \ge \int_a^b \tfrac12 f(c)\varphi_J = \tfrac12 f(c)(d_1 - c_1) > 0, which contradicts \int_a^b f = 0. Therefore f(x) = 0 for all x \in [a,b] (MAT-22 MCQ 23 option A). ∎
   Monotonicity 7.1.5(c) and Lemma 7.2.4 (T13).

Why continuity cannot be dropped (Ex 7.2.9): Define f(a) := 1 and f(x) := 0 for x \in (a,b]. Then f(x) \ge 0 on [a,b] and \int_a^b f = 0 by Theorem 7.1.3, yet f(a) = 1 \ne 0.

Marks: continuity neighborhood with f > \tfrac12 f(c) 1 · step-function lower bound \int_a^b f > 0 1 · counterexample

without continuity 1.

19 PYQ MTS-22 Q14

Leibniz rule for variable-limit integrals; derivatives of \int_0^{x^2}(1+t^3)^{-1}\,dt and \int_{x^2}^x \sqrt{1+t^2}\,dt.

KNOW FIRST

T23 T19 P11

Leibniz rule for variable limits (Bartle Ex 7.3.10; PYQ MTS-22 Q14): Let f : [a,b] \to \mathbb{R} be continuous and let u, v : [c,d] \to [a,b] be differentiable. If I(z) := \int_a^z f(t)\,dt, then I'(z) = f(z) by Theorem 7.3.6 (FTC 2). Since G(x) = \int_{u(x)}^{v(x)} f(t)\,dt = I(v(x)) - I(u(x)) by Theorem 7.2.13, the Chain Rule 6.1.6 gives G'(x) = I'(v(x))\,v'(x) - I'(u(x))\,u'(x) = f(v(x))\,v'(x) - f(u(x))\,u'(x).

(a) F(x) = \int_0^{x^2} (1 + t^3)^{-1}\,dt (Bartle Ex 7.3.11(a)): Here f(t) = (1 + t^3)^{-1} is continuous on [0,1], u(x) = 0, and v(x) = x^2 (v'(x) = 2x), so F'(x) = \frac{1}{1 + (x^2)^3}\cdot 2x = \frac{2x}{1 + x^6}.

(b) F(x) = \int_{x^2}^x \sqrt{1 + t^2}\,dt (Bartle Ex 7.3.11(b)): Here f(t) = \sqrt{1 + t^2} is continuous on [0,1], v(x) = x (v'=1), and u(x) = x^2 (u'=2x), so F'(x) = \sqrt{1 + x^2}\cdot 1 - \sqrt{1 + (x^2)^2}\cdot 2x = \sqrt{1 + x^2} - 2x\sqrt{1 + x^4}.

ANSWER **(a) F'(x) = \dfrac{2x}{1 + x^6}; (b) F'(x) = \sqrt{1 + x^2} - 2x\sqrt{1 + x^4}.**

Marks: general rule via FTC 2 + Chain Rule 1 · part (a) 1 · part (b) 1.

20

Mean Value Theorem for Integrals (Bartle Exercise 7.2.16).

KNOW FIRST

T15 T07 P07

PROOF (BARTLE EXERCISE 7.2.16)

1. Since f is continuous on the closed bounded interval [a,b], f \in \mathcal{R}[a,b] (Theorem 7.2.7) and by the Maximum–Minimum Theorem 5.3.4 (P07) f attains its minimum m := \min_{[a,b]} f and maximum M := \max_{[a,b]} f on [a,b].
   Theorem 7.2.7 and Max–Min Theorem 5.3.4 (P07).

2. Since m \le f(x) \le M for all x \in [a,b], Theorem 7.1.5(c) and Example 7.1.4(a) give $m(b - a) \le \int_a^b f \le M(b - a), \quad \text{so} \quad m \le \frac{1}{b - a}\int_a^b f \le M.$
   Monotonicity of the integral (7.1.5c) and division by b - a > 0.

3. By Bolzano's Intermediate Value Theorem 5.3.7 (P07), f attains every value between m and M, so there exists c \in [a,b] such that $f(c) = \frac{1}{b-a}\int_a^b f$, i.e. $\int_a^b f = f(c)(b - a)$. ∎
   Bolzano's IVT 5.3.7. (Alternative: apply the MVT 6.2.4 to F(x) = \int_a^x f, which gives c \in (a,b).)

Fails for step functions: On [0,2], g(x) = 1 on [0,1) and g(x) = 2 on [1,2] has \int_0^2 g = 3, but g(c)(2 - 0) = 3 \iff g(c) = 3/2, which g never attains.

Marks: m(b-a) \le \int_a^b f \le M(b-a) 1 · IVT conclusion 1 · step-function counterexample 1.

Section B · Paragraph / problem answers (Part 1: Q21–Q28)

6 marks each. Ceiling 72.

21

PYQ MAT-21 Q29 | also MTS-22 Q5, MTS-25 Q5, MAT-22 MCQ 14

Linearity \int_a^b (f+g) = \int_a^b f + \int_a^b g and monotonicity f \le g \Rightarrow \int_a^b f \le \int_a^b g (Theorem 7.1.5(b),(c)).

KNOW FIRST

T07 T03 P02 P05 Triangle inequality and the "arbitrary ε" rule M08

THEOREM 7.1.5(B), (C) (BARTLE P. 204)

Suppose f and g are in \mathcal{R}[a,b]. Then:
(b) The function f + g is in \mathcal{R}[a,b] and \int_a^b (f+g) = \int_a^b f + \int_a^b g.

(c) If f(x) \le g(x) for all x \in [a,b], then \int_a^b f \le \int_a^b g.

SKELETON

Check S(f+g;\dot P) = S(f;\dot P) + S(g;\dot P) and f \le g \Rightarrow S(f;\dot P) \le S(g;\dot P). Given \varepsilon > 0, choose \delta_\varepsilon = \min\{\delta_1,\delta_2\} so both sums are within \varepsilon/2 of their integrals. Use Triangle Inequality for (b) and sandwiching for (c).

PROOF OF THEOREM 7.1.5(B), (C) (BARTLE PP. 204–205)

1. For any tagged partition \dot P = \{([x_{i-1},x_i],t_i)\}_{i=1}^n of [a,b], properties of finite sums (P02) yield S(f+g;\dot P) = \sum_{i=1}^n \big(f(t_i) + g(t_i)\big)(x_i - x_{i-1}) = S(f;\dot P) + S(g;\dot P), and if f(x) \le g(x) on [a,b], then since x_i - x_{i-1} > 0 we have f(t_i)(x_i - x_{i-1}) \le g(t_i)(x_i x_{i-1}), so S(f;\dot P) \le S(g;\dot P).
   Additivity and order preservation of finite sums (P02).

2. Let \varepsilon > 0 be given. Since f, g \in \mathcal{R}[a,b], Definition 7.1.1 provides \delta_1 > 0 and \delta_2 > 0 such that \lVert P\rVert < \delta_1 \implies \left\lvert S(f;\dot P) - \int_a^b f\right\rvert < \frac{\varepsilon}{2}, \qquad \lVert P\rVert < \delta_2 \implies \left\lvert S(g;\dot P) - \int_a^b g\right\rvert < \frac{\varepsilon}{2}. Define \delta_\varepsilon := \min\{\delta_1, \delta_2\} > 0, and let \dot P be any tagged partition of [a,b] with \lVert P\rVert < \delta_\varepsilon.
   Definition 7.1.1 with tolerance \varepsilon/2 and common \delta_\varepsilon (M08).

3. **Proof of (b):** By Step 1, Step 2, and the Triangle Inequality, $\left\lvert S(f+g;\dot P) - \left(\int_a^b f + \int_a^b g\right)\right\rvert = \left\lvert\left(S(f;\dot P) \int_a^b f\right) + \left(S(g;\dot P) - \int_a^b g\right)\right\rvert \le \left\lvert S(f;\dot P) - \int_a^b f\right\rvert + \left\lvert S(g;\dot P) - \int_a^b g\right\rvert < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $f + g \in \mathcal{R}[a,b]$ and $\int_a^b (f+g) = \int_a^b f + \int_a^b g$.
   Triangle Inequality (P05) and Definition 7.1.1.

4. **Proof of (c):** For the same $\dot P$ with $\lVert P\rVert < \delta_\varepsilon$, unpacking the two absolute value inequalities of Step 2 gives $\int_a^b f - \frac{\varepsilon}{2} < S(f;\dot P) \quad \text{and} \quad S(g;\dot P) < \int_a^b g + \frac{\varepsilon}{2}$. Combining these with $S(f;\dot P) \le S(g;\dot P)$ from Step 1 gives $\int_a^b f \frac{\varepsilon}{2} < \int_a^b g + \frac{\varepsilon}{2}$, or $\int_a^b f < \int_a^b g + \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $\int_a^b f \le \int_a^b g$. ∎
   Arbitrary ε rule (P05).

Why strict inequality < need not hold (MAT-22 MCQ 14): Option (B) $\int_a^b f \le \int_a^b g$ is the

true statement; strict < fails whenever f = g, or when f \equiv 0 and g(a) = 1, g(x) = 0 on (a,b] (where \int_a^b f = 0 = \int_a^b g).

Marks: Riemann sum relations 1 · common \delta_\varepsilon setup 1 · proof of (b) 2 · proof of (c) and MCQ remark 2.

22

Thomae's function h on [0,1]: definition, proof that h \in \mathcal{R}[0,1] with \int_0^1 h = 0, and the set E for h(x) \ge 1/5.

KNOW FIRST

T09 T03 P14 M03 M10

DEFINITION OF THOMAE'S FUNCTION (BARTLE EXAMPLE 7.1.7, P. 206)

Define h : [0,1] \to \mathbb{R} by h(x) := 0 if x \in [0,1] is irrational, h(0) := 1, and h(x) := 1/n if x \in [0,1] is the rational number x = m/n where m, n ∈\in \mathbb{N} have no common integer factor except 1.

SKELETON

Given \varepsilon > 0, let E = \{x \in [0,1] : h(x) \ge \varepsilon/2\}, which has finitely many points n. Choose \delta = \varepsilon/(4n). Split \dot P into \dot P_1 (tags in E, at most 2n subintervals) and \dot P_2 (tags outside E, where h < \varepsilon/2).

PROOF THAT H \IN \MATHCAL{R}[0,1] AND \INT_0^1 H = 0 (BARTLE P. 206)

1. Let \varepsilon > 0 be given. Since h(m/n) = 1/n \ge \varepsilon/2 \iff n \le 2/\varepsilon, there are only finitely many denominators n \in \mathbb{N} and finitely many numerators m \in \{1,\dots,n\} satisfying 1/n \ge \varepsilon/2. Thus the set E := \{x \in [0,1] : h(x) \ge \varepsilon/2\} is a finite set.
   Archimedean property (P13).

2. Let n be the number of points in E, and define \delta := \dfrac{\varepsilon}{4n} > 0. Let \dot P be any tagged partition of [0,1] with \lVert P\rVert < \delta. Separate \dot P into \dot P_1 (the subset of subintervals with tags in E) and \dot P_2 (the subset with tags elsewhere in [0,1]).
   Tag-splitting tool (M10).

3. Allowing for a point of E serving as an endpoint tag of two adjacent subintervals, \dot P_1 has at most 2n subintervals, of total length at most 2n\delta = 2n\cdot\frac{\varepsilon}{4n} = \frac{\varepsilon}{2}. Since 0 < h(t_i) \le 1 on \dot P_1, we have S(h;\dot P_1) \le 1 \cdot 2n\delta = \frac{\varepsilon}{2}.
   Endpoint tag count (M03) and h(x) \le 1.

4. For tags t_i in \dot P_2, we have t_i \notin E, so 0 \le h(t_i) < \varepsilon/2. Since the total length of the subintervals in \dot P_2 is at most 1 (the length of [0,1]), we have S(h;\dot P_2) < \frac{\varepsilon}{2}\cdot 1 = \frac{\varepsilon}{2}.
   Height < \varepsilon/2 times total width \le 1 (M04).

5. Adding Steps 3 and 4 gives 0 \le S(h;\dot P) = S(h;\dot P_1) + S(h;\dot P_2) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. Since \varepsilon > 0 is arbitrary, h \in \mathcal{R}[0,1] and \int_0^1 h = 0. ∎
   Definition 7.1.1 with L = 0.

Explicit list of E = \{x \in [0,1] : h(x) \ge 1/5\}: We collect 0 (where h(0)=1) and all fractions m/n \in (0,1] in lowest terms with denominator n \in \{1,2,3,4,5\}:

ANSWER **E = \left\{0,\, 1,\, \frac12,\, \frac13,\, \frac23,\, \frac14,\, \frac34,\, \frac15,\, \frac25,\, \frac35,\, \frac45\right\} (11 points); \displaystyle\int_0^1 h = 0.**

Marks: definition of h 1 · finiteness of E and \delta = \varepsilon/(4n) 1.5 · estimates on \dot P_1 and \dot P_2 2.5 · list of 11 points in E 1.

23 PYQ MAT-21 Q8

State and prove the Cauchy Criterion for Riemann integrability (Theorem 7.2.1).

KNOW FIRST

T10 T03 P06

THEOREM 7.2.1 · CAUCHY CRITERION (BARTLE P. 208)

A function $f : [a,b] \to \mathbb{R}$ belongs to $\mathcal{R}[a,b]$ if and only if for every $\varepsilon > 0$ there exists $\eta_\varepsilon > 0$ such that if $\dot P$ and $\dot Q$ are any tagged partitions of $[a,b]$ with $\lVert P\rVert < \eta_\varepsilon$ and $\lVert Q\rVert < \eta_\varepsilon$, then $\lvert S(f;\dot P) - S(f;\dot Q)\rvert < \varepsilon$.

SKELETON

($\Rightarrow$) Put $\eta_\varepsilon = \delta_{\varepsilon/2}$ and use the Triangle Inequality. ($\Leftarrow$) Take decreasing $\delta_n > 0$ for $\varepsilon = 1/n$, choose $\dot P_n$ with $\lVert P_n\rVert < \delta_n$, show $(S(f;\dot P_m))$ is Cauchy in $\mathbb{R}$ with limit A, let $m \to \infty$ to get $\lvert S(f;\dot P_n) - A\rvert \le 1/n$, and pick $K > 2/\varepsilon$.

PROOF OF THEOREM 7.2.1 (BARTLE P. 208)

(\Rightarrow)

1. Suppose f \in \mathcal{R}[a,b] with integral L. Given \varepsilon > 0, let \eta_\varepsilon := \delta_{\varepsilon/2} > 0 from Definition 7.1.1. If \dot P, \dot Q have \lVert P\rVert < \eta_\varepsilon and \lVert Q\rVert < \eta_\varepsilon, then \lvert S(f;\dot P) - L\rvert < \varepsilon/2 and \lvert S(f;\dot Q) - L\rvert < \varepsilon/2, so \lvert S(f;\dot P) - S(f;\dot Q)\rvert \le \lvert S(f;\dot P) - L\rvert + \lvert L - S(f;\dot Q)\rvert < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.
   Definition 7.1.1 and Triangle Inequality (P05).

(\Leftarrow)

1. For each n \in \mathbb{N}, let \delta_n > 0 be the number \eta_{1/n} such that \lVert P\rVert, \lVert Q\rVert < \delta_n \Rightarrow \lvert S(f;\dot P) - S(f;\dot Q)\rvert < 1/n. By replacing \delta_n with \min\{\delta_1,\dots,\delta_n\} if necessary, we may assume \delta_n \ge \delta_{n+1} for all n \in \mathbb{N}.
   Hypothesis with \varepsilon = 1/n; forcing \delta_n decreasing ensures \lVert P_m\rVert < \delta_m \le \delta_n for m > n.

2. For each $n \in \mathbb{N}$, choose a tagged partition $\dot P_n$ with $\lVert P_n\rVert < \delta_n$. If $m > n$, then both $\dot P_m$ and $\dot P_n$ have norm $<\delta_n$, so $\lvert S(f;\dot P_m) - S(f;\dot P_n)\rvert < \frac{1}{n}$.
   Step 1 applied to $\dot P_m$ and $\dot P_n$.

3. Thus $(S(f;\dot P_m))_{m=1}^\infty$ is a Cauchy sequence in $\mathbb{R}$. By Theorem 3.5.5 (P06), it converges to a real number $A := \lim_m S(f;\dot P_m)$. Letting $m \to \infty$ in Step 2 gives $\lvert S(f;\dot P_n) - A\rvert \le 1/n$ for all $n \in \mathbb{N}$.
   Cauchy Convergence Criterion for sequences in $\mathbb{R}$ (P06).

4. Given $\varepsilon > 0$, choose $K \in \mathbb{N}$ with $K > 2/\varepsilon$. For any tagged partition $\dot Q$ with $\lVert Q\rVert < \delta_K$, $\lvert S(f;\dot Q) - A\rvert \le \lvert S(f;\dot Q) - S(f;\dot P_K)\rvert + \lvert S(f;\dot P_K) - A\rvert < \frac{1}{K} + \frac{1}{K} = \frac{2}{K} < \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $f \in \mathcal{R}[a,b]$ with $\int_a^b f = A$. ∎
   Triangle Inequality and Definition 7.1.1 with $\delta_\varepsilon = \delta_K$.

Marks: statement 1 · ($\Rightarrow$) proof 1.5 · ($\Leftarrow$) Cauchy sequence & limit A 2 · final $2/K < \varepsilon$ verification 1.5.

24 PYQ MAT-22 Q4 also MAT-22 MCQ 17, MAT-21 MCQ 12, 16 · adapted

Dirichlet's function and h(x) = x + 1 on \mathbb{Q}\cap[0,1], 0 on irrationals, are not in \mathcal{R}[0,1].

KNOW FIRST

T11 T10 P14

SKELETON

Set \varepsilon_0 = 1/2 (or 1). For any \eta > 0, pick a partition P of [0,1] with \lVert P\rVert < \eta; tag P with rationals to get \dot P and with irrationals to get \dot Q, and show \lvert S(\dot P) - S(\dot Q)\rvert \ge \varepsilon_0.

PROOFS (BARTLE EXAMPLE 7.2.2(B) & EXERCISE 7.2.2)

Part 1: Dirichlet's function f(x) = 1 (x \in \mathbb{Q}\cap[0,1]), 0 (x \in [0,1]\setminus\mathbb{Q}) C7.

1. Put \varepsilon_0 := \tfrac12 > 0, and let \eta > 0 be arbitrary. Choose any partition P = \{[x_{i-1},x_i]\}_{i=1}^n of [0,1] with \lVert P\rVert < \eta.
   Set \varepsilon_0 = 1/2 to negate Theorem 7.2.1.
2. Since \mathbb{Q} and \mathbb{R}\setminus\mathbb{Q} are both dense in \mathbb{R}, each subinterval [x_{i-1},x_i] contains a rational point t_i and an irrational point s_i. Let \dot P be P tagged with the rational points t_i, and let \dot Q be P tagged with the irrational points s_i.
   Density of rationals and irrationals (P14).
3. Then S(f;\dot P) = \sum_{i=1}^n 1\cdot(x_i - x_{i-1}) = 1 and S(f;\dot Q) = \sum_{i=1}^n 0\cdot(x_i x_{i-1}) = 0, so \lvert S(f;\dot P) - S(f;\dot Q)\rvert = \lvert 1 - 0\rvert = 1 \ge \frac12 = \varepsilon_0. Since \eta > 0 was arbitrary, the Cauchy Criterion 7.2.1 fails and f \notin \mathcal{R}[0,1] (MAT-22 MCQ 17 & MAT-21 MCQ 12 option B).
   Negation of Theorem 7.2.1.

Part 2: h(x) = x + 1 (x \in \mathbb{Q}\cap[0,1]), 0 (x \in

[0,1]\setminus\mathbb{Q}) (Bartle Ex 7.2.2; MAT-21 MCQ 16).

1. Put \varepsilon_0 := 1 > 0. For any \eta > 0 and the same \dot P (rational tags t_i \in [0,1]) and \dot Q (irrational tags s_i), we have h(t_i) = t_i + 1 \ge 1 and h(s_i) = 0, so S(h;\dot P) = \sum_{i=1}^n (t_i + 1)(x_i - x_{i-1}) \ge \sum_{i=1}^n 1\cdot(x_i - x_{i-1}) = 1, \qquad S(h;\dot Q) = 0. Hence \lvert S(h;\dot P) - S(h;\dot Q)\rvert \ge 1 = \varepsilon_0, so h \notin \mathcal{R}[0,1] (MAT-21 MCQ 16 option B). ∎
   t_i \ge 0 \Rightarrow t_i + 1 \ge 1 on [0,1].

Marks: Cauchy negation setup 1 · Dirichlet proof 2.5 · h(x)=x+1 proof 2.5.

25 PYQ MTS-26 Q18 also MTS-22 Q6, MAT-21 Q7

State and prove the Squeeze Theorem (7.2.3); apply it to h(x) = x on [0,1] (Example 7.2.6(b)).

KNOW FIRST

T12 T10 T14 P03

THEOREM 7.2.3 · SQUEEZE THEOREM (BARTLE P. 209)

Let $f : [a,b] \to \mathbb{R}$. Then $f \in \mathcal{R}[a,b]$ if and only if for every $\varepsilon > 0$ there exist functions α_ε and ω_ε in $\mathcal{R}[a,b]$ with $\alpha_\varepsilon(x) \le f(x) \le \omega_\varepsilon(x) \quad \text{for all } x \in [a,b] \qquad \text{and} \qquad \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) < \varepsilon$.

PROOF OF THEOREM 7.2.3 & EXAMPLE 7.2.6(B) (BARTLE PP. 209–211)

Proof of 7.2.3: (\Rightarrow) If $f \in \mathcal{R}[a,b]$, take $\alpha_\varepsilon = \omega_\varepsilon = f$; then $\int_a^b(\omega_\varepsilon - \alpha_\varepsilon) = 0 < \varepsilon$. **(\Leftarrow)**

1. Let $\varepsilon > 0$. Since $\alpha_\varepsilon, \omega_\varepsilon \in \mathcal{R}[a,b]$, there exists $\delta_\varepsilon > 0$ such that any tagged partition $\dot P$ with $\lVert P\rVert < \delta_\varepsilon$ satisfies $\big\lvert S(\alpha_\varepsilon;\dot P) - \int_a^b \alpha_\varepsilon\big\rvert < \varepsilon$ and $\big\lvert S(\omega_\varepsilon;\dot P) - \int_a^b \omega_\varepsilon\big\rvert < \varepsilon$.

   Definition 7.1.1 applied to α_ε and ω_ε, taking $\min$ of the two deltas (M08).

2. Since \alpha_\varepsilon \le f \le \omega_\varepsilon, we have S(\alpha_\varepsilon;\dot P) \le S(f;\dot P) \le S(\omega_\varepsilon;\dot P), and therefore \int_a^b \alpha_\varepsilon - \varepsilon < S(f;\dot P) < \int_a^b \omega_\varepsilon + \varepsilon. If \dot Q is any other tagged partition with \lVert Q\rVert < \delta_\varepsilon, then \int_a^b \alpha_\varepsilon - \varepsilon < S(f;\dot Q) < \int_a^b \omega_\varepsilon + \varepsilon as well.
   Order preservation of Riemann sums (P02).

3. Subtracting these two inequalities and using Theorem 7.1.5 gives \lvert S(f;\dot P) - S(f;\dot Q)\rvert < \int_a^b \omega_\varepsilon - \int_a^b \alpha_\varepsilon + 2\varepsilon = \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) + 2\varepsilon < \varepsilon + 2\varepsilon = 3\varepsilon. Since \varepsilon > 0 is arbitrary, the Cauchy Criterion 7.2.1 implies f \in \mathcal{R}[a,b].
   Theorem 7.1.5 and Cauchy Criterion 7.2.1 (T10).

Application to h(x) = x on [0,1] (Example 7.2.6(b)):

1. On the n-part equal partition $P_n = (0, \tfrac1n, \tfrac2n, \dots, 1)$, define step functions $\alpha_n(x) := \tfrac{k-1}{n}$ and $\omega_n(x) := \tfrac{k}{n}$ for $x \in [\tfrac{k-1}{n}, \tfrac{k}{n})$ $(k=1,\dots,n-1$, and closed on $[\tfrac{n-1}{n},1])$. Then $\alpha_n \le h \le \omega_n$ and by Theorem 7.2.5, $\int_0^1 \alpha_n = \frac{1}{n}\sum_{k=1}^n \frac{k-1}{n} = \frac{(n-1)n}{2n^2} = \frac12\left(1 \frac1n\right)$, $\quad \int_0^1 \omega_n = \frac{1}{n}\sum_{k=1}^n \frac{k}{n} = \frac{n(n+1)}{2n^2} = \frac12\left(1 + \frac1n\right)$. Thus $\int_0^1(\omega_n - \alpha_n) = \tfrac1n < \varepsilon$ for $n > 1/\varepsilon$, so $h \in \mathcal{R}[0,1]$ by 7.2.3, and $\tfrac12(1-\tfrac1n) \le \int_0^1 x\,dx \le \tfrac12(1+\tfrac1n)\ \forall\, n \implies \int_0^1 x\,dx = \tfrac12$. ∎

   Sum of first n integers (P03) and Theorem 7.2.3.

ANSWER $\displaystyle\int_0^1 x\,dx = \frac12.$

Marks: statement of 7.2.3 1 · proof of 7.2.3 3 · step functions α_n, ω_n and $\int_0^1 x\,dx = 1/2$ 2.

26 PYQ MTS-26 Q5 also MAT-22 Q16

Every step function is integrable (Theorem 7.2.5); evaluate \int_0^3 g.

KNOW FIRST

T14 T13 T07 M06

THEOREM 7.2.5 · INTEGRABILITY OF STEP FUNCTIONS (BARTLE P. 210)

If \varphi : [a,b] \to \mathbb{R} is a step function, then \varphi \in \mathcal{R}[a,b]. If \varphi = \sum_{j=1}^m k_j\varphi_{J_j} where J_j has endpoints c_j < d_j, then \int_a^b \varphi = \sum_{j=1}^m k_j(d_j - c_j).

PROOF OF THEOREM 7.2.5 (BARTLE PP. 210–211)

1. First, by Lemma 7.2.4, for any subinterval $J \subseteq [a,b]$ with endpoints $c < d$, the elementary step function $\varphi_J(x) = 1$ on J and 0 elsewhere belongs to $\mathcal{R}[a,b]$ with $\int_a^b \varphi_J = d - c$ (for $J = [c,d]$, $\lVert P\rVert < \delta \Rightarrow (d-c)-2\delta \le S(\varphi_{[c,d]};\dot P) \le (d-c)+2\delta$, so $\delta_\varepsilon = \varepsilon/4$ works; the other three interval types differ at ≤ 2 endpoints, so Theorem 7.1.3 applies).
   Lemma 7.2.4 (T13).

2. By Definition 5.4.9, if $\varphi : [a,b] \to \mathbb{R}$ is an arbitrary step function, $[a,b]$ is the union of finitely many non-overlapping subintervals $J_1, \dots, J_m$ with endpoints $c_j \le d_j$ such that φ has constant value k_j on J_j. Thus φ can be written as the linear combination $\varphi = \sum_{j=1}^m k_j\varphi_{J_j}$.
   Exercise 7.2.5 / M06.

3 By Lemma 7.2.4 and linearity of the integral (Theorem 7.1.5(a),(b), applied inductively to m terms), \varphi \in \mathcal{R}[a,b] and \int_a^b \varphi = \sum_{j=1}^m k_j\int_a^b \varphi_{J_j} = \sum_{j=1}^m k_j(d_j - c_j). \quad \blacksquare

Theorem 7.1.5(a),(b) (T07).

Example 7.2.6(a): For g(x) = 2 on [0,1] and g(x) = 3 on (1,3], we have g = 2\varphi_{[0,1]} + 3\varphi_{(1,3]}, so \int_0^3 g = 2(1 - 0) + 3(3 - 1) = 2 + 6 = 8.

ANSWER **\displaystyle\int_a^b \varphi = \sum_{j=1}^m k_j(d_j - c_j) and \displaystyle\int_0^3 □g = 8.**

Marks: Lemma 7.2.4 proof/statement 2 · decomposition \varphi = \sum k_j\varphi_{J_j} 1.5 · linearity & formula 1.5 · \int_0^3 g = 8 1.

27 PYQ MAT-22 Q37b

Restriction to [c,d] \subseteq [a,b] (Corollary 7.2.10) and subdivision formula (Corollary 7.2.11).

KNOW FIRST

T18 T17

COROLLARIES 7.2.10 & 7.2.11 (BARTLE P. 214)

If $f \in \mathcal{R}[a,b]$ and $[c,d] \subseteq [a,b]$, then $f|_{[c,d]} \in \mathcal{R}[c,d]$. If $a = c_0 < c_1 < \cdots < c_m = b$, then $\int_a^b f = \sum_{i=1}^m \int_{c_{i-1}}^{c_i} f$.

PROOFS

1. **Proof of Corollary 7.2.10:** Let f \in \mathcal{R}[a,b] and a \le c < d \le b. If c = a, then f|_{[c,b]} = f \in \mathcal{R}[c,b] by hypothesis; if c > a, then c \in (a,b), so the Additivity Theorem 7.2.9 (\Rightarrow) implies that the restriction f|_{[c,b]} belongs to \mathcal{R}[c,b].
   Additivity Theorem 7.2.9 (\Rightarrow) at the cut point c \in (a,b).

2. Next, consider f|_{[c,b]} \in \mathcal{R}[c,b]. If d = b, then f|_{[c,d]} = f|_{[c,b]} \in \mathcal{R}[c,d]; if d < b, then d \in (c,b), so applying Theorem 7.2.9 (\Rightarrow) to f|_{[c,b]} on [c,b] at the interior point d shows that its restriction f|_{[c,d]} belongs to \mathcal{R}[c,d].
   Second application of Theorem 7.2.9 (\Rightarrow) at d \in (c,b).

3. **Proof of Corollary 7.2.11 (by induction on m):** By Corollary 7.2.10, each restriction f|_{[c_{i-1},c_i]} belongs to \mathcal{R}[c_{i-1},c_i]. For m = 1, \int_a^b f = \int_{c_0}^{c_1} f is immediate; for m = 2, \int_a^b f = \int_{c_0}^{c_1} f + \int_{c_1}^{c_2} f is Theorem 7.2.9. Assume the formula holds for m - 1 subintervals. Given a = c_0 < c_1 < \cdots < c_{m-1} < c_m = b, apply Theorem 7.2.9 at c = c_{m-1} \in (a,b) to get \int_a^b f = \int_a^{c_{m-1}} f + \int_{c_{m-1}}^{c_m} f = \sum_{i=1}^{m-1} \int_{c_{i-1}}^{c_i} f + \int_{c_{m-1}}^{c_m} f = \sum_{i=1}^m \int_{c_{i-1}}^{c_i} f. \quad \blacksquare
   Induction hypothesis applied on [a,c_{m-1}].

Marks: split at c 1.5 · split at d 1.5 · induction base & step for 7.2.11 3.

28

PYQ MAT-22 MCQ 24, 25 · adapted

also MAT-21 MCQ 18 · adapted

Bounded f integrable on every [c,b] (c \in (a,b)) is in \mathcal{R}[a,b] (Bartle Ex 7.2.11–7.2.13); application to \sin(1/x).

KNOW FIRST

T18 T12 T17 T08

PROOF OF BARTLE EXERCISES 7.2.11–7.2.13

1. Assume \lvert f(x)\rvert \le M on [a,b] (M > 0) and f|_{[c,b]} \in \mathcal{R}[c,b] for every c \in (a,b). Given \varepsilon > 0, choose c \in (a,b) such that c - a < \dfrac{\varepsilon}{4M}.
   Make the trouble strip [a,c] near a narrower than \varepsilon/(4M).

2. Define \alpha_c, \omega_c : [a,b] \to \mathbb{R} by \alpha_c(x) := -M and \omega_c(x) := M for x \in [a,c), and \alpha_c(x) := \omega_c(x) := f(x) for x \in [c,b]. Since \alpha_c, \omega_c are constant on [a,c) and equal to f \in \mathcal{R}[c,b] on [c,b], Theorem 7.2.9 (\Leftarrow) and Theorem 7.1.3 imply \alpha_c, \omega_c \in \mathcal{R}[a,b] and \alpha_c(x) \le f(x) \le \omega_c(x) on [a,b].
   Additivity Theorem 7.2.9 (\Leftarrow) glues the constant piece on [a,c] to f on [c,b].

3. Since \omega_c - \alpha_c = 2M on [a,c) and 0 on [c,b], we have \int_a^b (\omega_c - \alpha_c) = \int_a^c 2M + \int_c^b 0 = 2M(c - a) < 2M\cdot\frac{\varepsilon}{4M} = \frac{\varepsilon}{2} < \varepsilon. By the Squeeze Theorem 7.2.3, f \in \mathcal{R}[a,b].
   Squeeze Theorem 7.2.3 (T12).

4. Furthermore, for any c \in (a,b), Theorem 7.2.9 gives \int_a^b f - \int_c^b f = \int_a^c f, so \left\lvert\int_a^b f - \int_c^b f\right\rvert = \left\lvert\int_a^c f\right\rvert \le M(c - a) \to 0 as c \to a^+, proving \lim_{c\to a^+} \int_c^b f = \int_a^b f. (Thus in MAT-22 MCQ 24 / MAT-21 MCQ 18, the statement that does **not** hold is **(B) f \notin \mathcal{R}[a,b]**.)
   Exercise 7.1.8 bound \lvert\int_a^c f\rvert \le M(c-a) and Squeeze Theorem for limits.

5. **Application to g(x) = \sin(1/x) (x \in (0,1]), g(0) = 0 (Ex 7.2.12, MAT-22 MCQ 25):** We have \lvert g(x)\rvert \le 1 on [0,1], and for each c \in (0,1), g is continuous on [c,1] and hence in \mathcal{R}[c,1] by Theorem 7.2.7. By Step 3, g \in \mathcal{R}[0,1] (MCQ 25 option A).
   Apply Exercise 7.2.11 with M = 1.

6. **Why boundedness cannot be dropped (Ex 7.2.13):** f(x) = 1/x on (0,1], f(0) = 0 C16 is continuous on [c,1] (so f \in \mathcal{R}[c,1]) for every c \in (0,1), but f is unbounded on [0,1], so f \notin \mathcal{R}[0,1] by Theorem 7.1.6. ∎
   Counterexample C16.

Marks: \alpha_c, \omega_c construction & Squeeze 2.5 · limit \int_c^b f \to \int_a^b f 1.5 · \sin(1/x) \in \mathcal{R}[0,1] 1 · 1/x counterexample 1.

Section B · Paragraph / problem answers (Part 2: Q29–Q36)
6 marks each. Ceiling 72.

29 PYQ MTS-26 Q8

State and prove the First Form of the Fundamental Theorem of Calculus (Theorem 7.3.1).

KNOW FIRST
T20 T03 T18 P12 M05

THEOREM 7.3.1 · FUNDAMENTAL THEOREM OF CALCULUS, FIRST FORM (BARTLE P. 216)

Suppose there is a finite set E in [a,b] and functions f, F : [a,b] \to \mathbb{R} such that:
(a) F is continuous on [a,b];

(b) F'(x) = f(x) for all x \in [a,b] \setminus E;

(c) f belongs to \mathcal{R}[a,b].

Then we have \int_a^b f = F(b) - F(a).

SKELETON

For E = \{a,b\}, given \varepsilon > 0 choose \delta_\varepsilon from f \in \mathcal{R}[a,b] and a partition P with \lVert P\rVert < \delta_\varepsilon. Apply the MVT 6.2.4 to F on each [x_{i-1},x_i] to get F(x_i) F(x_{i-1}) = f(u_i) (x_i - x_{i-1}); sum to get F(b) - F(a) = S(f;\dot P_u), which is within \varepsilon of \int_a^b f.

PROOF OF THEOREM 7.3.1 (BARTLE PP. 216–217)

1. We first prove the case where E = \{a,b\}. Let \varepsilon > 0 be given. Since f \in \mathcal{R}[a,b] by hypothesis (c), there exists \delta_\varepsilon > 0 such that if \dot P is any tagged partition of [a,b] with \lVert P\rVert < \delta_\varepsilon, then \left\lvert S(f;\dot P) - \int_a^b f\right\rvert < \varepsilon.
   Definition 7.1.1 (T03).

2. Let P = \{[x_{i-1},x_i]\}_{i=1}^n be a partition of [a,b] with \lVert P\rVert < \delta_\varepsilon. Since F is continuous on [x_{i-1},x_i] by (a) and differentiable on (x_{i-1},x_i) \subseteq [a,b]\setminus E by (b), the Mean Value Theorem 6.2.4 (P12) applied to F on [x_{i-1},x_i] implies there exists u_i \in (x_{i-1},x_i) such that F(x_i) - F(x_{i-1}) = F'(u_i)(x_i - x_{i-1}) = f(u_i)(x_i - x_{i-1}) \quad \text{for } i = 1, \dots, n.
   Mean Value Theorem 6.2.4 on [x_{i-1},x_i], using u_i \notin E \Rightarrow F'(u_i) = f(u_i).

3. Adding these n equations and telescoping the left-hand side gives F(b) - F(a) = \sum_{i=1}^n \big(F(x_i) - F(x_{i-1})\big) = \sum_{i=1}^n f(u_i)(x_i - x_{i-1}) = S(f;\dot P_u), where \dot P_u := \{([x_{i-1},x_i],u_i)\}_{i=1}^n is a tagged partition of [a,b] with \lVert P\rVert < \delta_\varepsilon.
   Telescoping sum (M01, M05).

4. Substituting $S(f;\dot P_u) = F(b) - F(a)$ into Step 1 gives $\big\lvert F(b) - F(a) \int_a^b f\big\rvert < \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $\int_a^b f = F(b) - F(a)$.
Arbitrary ε rule (P05).

5. **General finite set E (Exercise 7.3.1):** Let $a = c_0 < c_1 < \cdots < c_m = b$ be the points of $E \cup \{a,b\}$ in increasing order. On each $[c_{k-1},c_k]$, F is continuous, $F'(x) = f(x)$ on (c_{k-1},c_k), and $f|_{[c_{k-1},c_k]} \in \mathcal{R}[c_{k-1},c_k]$ by Corollary 7.2.10. Applying the first part on each subinterval and summing via Corollary 7.2.11 gives $\int_a^b f = \sum_{k=1}^m \big(F(c_k) F(c_{k-1})\big) = F(b) - F(a)$. ∎
Corollaries 7.2.10 and 7.2.11 (T18).

Marks: statement with (a),(b),(c) 1.5 · MVT step on $[x_{i-1},x_i]$ 2 · telescoping to $S(f;\dot P_u)$ and conclusion 1.5 · general finite E 1.

30 PYQ MTS-25 Q6 also MAT-22 Q18, MAT-21 Q19

Substitution Theorem 7.3.8: statement, proof, five substitution evaluations, and $\int_0^4 \frac{\sin\sqrt t}{\sqrt t}\,dt$ via FTC 1.

KNOW FIRST

T25 T23 T20 T18 P11

THEOREM 7.3.8 · SUBSTITUTION THEOREM (BARTLE P. 220)

Let $J := [\alpha,\beta]$ and let $\varphi : J \to \mathbb{R}$ have a continuous derivative on J. If $f : I \to \mathbb{R}$ is continuous on an interval I containing $\varphi(J)$, then $\int_\alpha^\beta f(\varphi(t))\,\varphi'(t)\,dt = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)\,dx$.

PROOF OF THEOREM 7.3.8 (BARTLE EXERCISE 7.3.17, P. 224)

1. Define F(u) := \int_{\varphi(\alpha)}^u f(x)\,dx for u \in I. Since f is continuous on I, Theorem 7.3.6 (FTC 2) implies that F'(u) = f(u) for all u \in I.
   Theorem 7.3.6 (T23).

2. Define H(t) := F(\varphi(t)) for t \in J = [\alpha,\beta]. By the Chain Rule 6.1.6, H is differentiable on J and H'(t) = F'(\varphi(t))\varphi'(t) = f(\varphi(t))\varphi'(t) for all t \in J.
   Chain Rule 6.1.6 (P11).

3. Since f, \varphi, and \varphi' are continuous, (f\circ\varphi)\varphi' is continuous on [\alpha,\beta] and hence in \mathcal{R}[\alpha,\beta] (Theorem 7.2.7). Since H(\alpha) = F(\varphi(\alpha)) = 0, Theorem 7.3.1 (FTC 1 with E = \varnothing) gives \int_\alpha^\beta f(\varphi(t))\,\varphi'(t)\,dt = H(\beta) - H(\alpha) = F(\varphi(\beta)) = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)\,dx. \quad \blacksquare
   Theorem 7.2.7 and Theorem 7.3.1 (T20).

Five substitution evaluations (Example 7.3.9(a) & Exercise 7.3.18):

- **1. \int_1^4 \frac{\sin\sqrt t}{\sqrt t}\,dt (MTS-25 Q6, MAT-21 Q19):** Put \varphi(t) = \sqrt t on

[1,4] ($\varphi'(t) = \frac{1}{2\sqrt t}$ continuous on [1,4]) and $f(x) = 2\sin x$: $\int_1^4 \frac{\sin\sqrt t}{\sqrt t}\,dt = \int_1^2 2\sin x\,dx = \big[-2\cos x\big]_1^2 = 2(\cos 1 - \cos 2) \approx 1.912898$.

- **2. $\int_1^4 \frac{\cos\sqrt t}{\sqrt t}\,dt$ (Ex 7.3.18(d)):** Put $\varphi(t) = \sqrt t$ on [1,4] and $f(x) = 2\cos x$: $\int_1^4 \frac{\cos\sqrt t}{\sqrt t}\,dt = \int_1^2 2\cos x\,dx = \big[2\sin x\big]_1^2 = 2(\sin 2 - \sin 1) \approx 0.135653$.
- **3. $\int_0^1 t\sqrt{1+t^2}\,dt$ (Ex 7.3.18(a)):** Put $\varphi(t) = 1 + t^2$ on [0,1] ($\varphi'(t) = 2t$ continuous) and $f(x) = \tfrac12\sqrt x$: $\int_0^1 t\sqrt{1+t^2}\,dt = \int_1^2 \frac12 x^{1/2}\,dx = \left[\frac13 x^{3/2}\right]_1^2 = \frac{2\sqrt 2 - 1}{3} \approx 0.609476$.
- **4. $\int_0^2 t^2(1+t^3)^{-1/2}\,dt$ (Ex 7.3.18(b)):** Put $\varphi(t) = 1 + t^3$ on [0,2] ($\varphi'(t) = 3t^2$ continuous) and $f(x) = \tfrac13 x^{-1/2}$: $\int_0^2 t^2(1+t^3)^{-1/2}\,dt = \int_1^9 \frac13 x^{-1/2}\,dx = \left[\frac23\sqrt x\right]_1^9 = \frac23(3 - 1) = \frac{4}{3}$.
- **5. $\int_1^4 \frac{\sqrt{1+\sqrt t}}{\sqrt t}\,dt$ (Ex 7.3.18(c)):** Put $\varphi(t) = 1 + \sqrt t$ on [1,4] ($\varphi'(t) = \frac{1}{2\sqrt t}$ continuous on [1,4]) and $f(x) = 2\sqrt x$: $\int_1^4 \frac{\sqrt{1+\sqrt t}}{\sqrt t}\,dt = \int_2^3 2x^{1/2}\,dx = \left[\frac43 x^{3/2}\right]_2^3 = \frac43(3\sqrt 3 - 2\sqrt 2) \approx 3.156967$.

Why 7.3.8 fails on [0,4] and how $\int_0^4 \frac{\sin\sqrt t}{\sqrt t}\,dt$ is evaluated (Example 7.3.9(b); PYQ MAT-22 Q18): On [0,4],

\varphi(t) = \sqrt t has \varphi'(t) = \frac{1}{2\sqrt t}, which is undefined and unbounded at t = 0, so Theorem 7.3.8 does not apply. Instead, let g(t) := \frac{\sin\sqrt t}{\sqrt t} on (0,4] and g(0) := 1. Since \lvert\sin u\rvert \le \lvert u\rvert, \lvert g(t)\rvert \le 1 is bounded on [0,4] and continuous on [c,4] for every c \in (0,4), so g \in \mathcal{R}[0,4] by Exercise 7.2.11 (Q28). Define F(t) := -2\cos\sqrt t on [0,4]; then F is continuous on [0,4] and F'(t) = g(t) for all t \in (0,4] (E = \{0\}). By FTC 1 (Theorem 7.3.1), \int_0^4 \frac{\sin\sqrt t}{\sqrt t}\,dt = F(4) - F(0) = -2\cos 2 - (-2\cos 0) = 2(1 - \cos 2) \approx 2.832294.

ANSWER **\int_1^4 \frac{\sin\sqrt t}{\sqrt t}\,dt = 2(\cos 1 - \cos 2) \approx 1.912898;\quad \int_0^4 \frac{\sin\sqrt t}{\sqrt t}\,dt = 2(1 - \cos 2) \approx 2.832294.**

Marks: statement & proof of 7.3.8 2 · substitution evaluations 2 · [0,4] explanation & FTC 1 evaluation 2(1-\cos 2) 2.

31 PYQ MAT-22 Q6

State Lebesgue's Integrability Criterion (7.3.12) and apply it to five benchmark functions (Examples 7.3.13(a)–(e)).

KNOW FIRST

T27 T26 T08

THEOREM 7.3.12 · LEBESGUE'S INTEGRABILITY CRITERION (BARTLE P. 221)

A bounded function f : [a,b] \to \mathbb{R} is Riemann integrable if and only if it is continuous almost everywhere on [a,b] (equivalently, the set D(f) of points of discontinuity of f on [a,b] is a null set).

SKELETON

For each function on [0,1], check (1) whether it is bounded, and (2) whether its discontinuity set D is a null set.

Applications (Bartle Examples 7.3.13(a)–(e)):

- **(a) Any step function \varphi on [0,1]:** \varphi takes finitely many values, so it is bounded, and it is discontinuous at most at the finitely many endpoints of its constant intervals. Since every finite set is a null set, \varphi \in \mathcal{R}[0,1].
- **(b) Any monotone function f on [0,1]:** f is bounded between f(0) and f(1), and by Theorem 5.6.4 (P10) its set of discontinuities is at most

countable, hence a null set (T26). By Theorem 7.3.12, f \in \mathcal{R}[0,1].

- **(c) G(1/n) = 1/n (n \in \mathbb{N}), G(x) = 0 elsewhere on [0,1] (Example 7.1.4(d)):** G is bounded (0 \le G(x) \le 1), continuous at x = 0 (since \lim_{x\to 0} G(x) = 0 = G(0)) and at every point outside \{1/n : n\in\mathbb{N}\}, and discontinuous precisely on the countable set D = \{1, \tfrac12, \tfrac13, \dots, \tfrac1n, \dots\}. Since every countable set is a null set, G \in \mathcal{R}[0,1].
- **(d) Dirichlet's function f on [0,1]** C7 **:** f is bounded (0 \le f \le 1), but f is discontinuous at **every** point of [0,1]. Since D = [0,1] is not a null set, f \notin \mathcal{R}[0,1].
- **(e) Thomae's function h on [0,1]** C8 **:** h is bounded (0 \le h \le 1), continuous at every irrational in [0,1], and discontinuous precisely on D = \mathbb{Q} \cap [0,1]. By Example 7.3.11, \mathbb{Q}\cap[0,1] is a null set, so h \in \mathcal{R}[0,1].

Marks: statement of Theorem 7.3.12 1 · 1 mark for each of the five cases (a)–(e).

32 PYQ MAT-21 Q4

State and prove the Composition Theorem (7.3.14); counterexample when \varphi is not continuous.

KNOW FIRST

T28 T27 T26 P07

THEOREM 7.3.14 · COMPOSITION THEOREM (BARTLE P. 222)

Let $f \in \mathcal{R}[a,b]$ with $f([a,b]) \subseteq [c,d]$, and let $\varphi : [c,d] \to \mathbb{R}$ be continuous. Then the composition $\varphi \circ f$ belongs to $\mathcal{R}[a,b]$.

PROOF OF THEOREM 7.3.14 (BARTLE P. 222)

1. If f is continuous at a point u \in [a,b], then since \varphi is continuous at f(u) \in [c,d], the composite function \varphi \circ f is also continuous at u.
   Composition of continuous functions is continuous (P07).

2. Consequently, the set D of points of discontinuity of \varphi \circ f on [a,b] is contained in the set D(f) of points of discontinuity of f.
   Contrapositive of Step 1: u \in D \Rightarrow u \in D(f).

3. Since f \in \mathcal{R}[a,b], Lebesgue's Integrability Criterion 7.3.12 implies that D(f) is a null set. Since any subset of a null set is a null set (T26), D is a null set.
   Theorem 7.3.12 (\Rightarrow) and subset property of null sets.

4. Moreover, since \varphi is continuous on the closed bounded interval [c,d], \varphi is bounded on [c,d] (Theorem 5.3.2), so \varphi \circ f is bounded on [a,b]. By Lebesgue's Integrability Criterion 7.3.12, \varphi \circ f \in \mathcal{R}[a,b]. ∎
   Boundedness Theorem 5.3.2 (P07) and Theorem 7.3.12 (\Leftarrow).

Counterexample showing continuity of \varphi cannot be dropped (Exercise 7.3.22): Let h \in \mathcal{R}[0,1] be Thomae's function C8, and

define \varphi : [0,1] \to \mathbb{R} by \varphi(0) := 0 and \varphi(y) := 1 for y \in (0,1]. Note that \varphi \in \mathcal{R}[0,1] (step function, discontinuous only at 0). However, for x \in [0,1], (\varphi \circ h)(x) = 1 if x \in \mathbb{Q}\cap[0,1] (where h(x) = 1/n > 0) and (\varphi \circ h)(x) = \varphi(0) = 0 if x \in [0,1]\setminus\mathbb{Q}: thus \varphi \circ h is Dirichlet's function C7, which is **not** in \mathcal{R}[0,1].

Marks: statement 1 · D(\varphi\circ f) \subseteq D(f) null 2 · boundedness & Lebesgue conclusion 1.5 · counterexample 1.5.

33 PYQ MTS-25 Q8 also MTS-22 Q7, MAT-21 Q20

If f \in \mathcal{R}[a,b], then \lvert f\rvert \in \mathcal{R}[a,b] and \big\lvert\int_a^b f\big\rvert \le \int_a^b \lvert f\rvert \le M(b-a) (Corollary 7.3.15); converse false.

KNOW FIRST

T29 T28 T08 T07

COROLLARY 7.3.15 (BARTLE P. 222)

Suppose f \in \mathcal{R}[a,b]. Then its absolute value \lvert f\rvert is in \mathcal{R}[a,b], and \left\lvert\int_a^b f\right\rvert \le \int_a^b \lvert f\rvert \le M(b - a), where \lvert f(x)\rvert \le M for all x \in [a,b].

PROOF OF COROLLARY 7.3.15 (BARTLE P. 222)

1. Since f \in \mathcal{R}[a,b], Theorem 7.1.6 implies that f is bounded on [a,b], say \lvert f(x)\rvert \le M for all x \in [a,b], so f([a,b]) \subseteq [-M,M].
   Boundedness Theorem 7.1.6 (T08).

2. The function \varphi(t) := \lvert t\rvert is continuous on [-M,M]. By the Composition Theorem 7.3.14, \lvert f\rvert = \varphi \circ f belongs to \mathcal{R}[a,b].
   Composition Theorem 7.3.14 (T28).

3. Since -\lvert f(x)\rvert \le f(x) \le \lvert f(x)\rvert for all x \in [a,b], Theorem 7.1.5(a) and (c) give -\int_a^b \lvert f\rvert \le \int_a^b f \le \int_a^b \lvert f\rvert, \quad \text{and hence} \quad \left\lvert\int_a^b f\right\rvert \le \int_a^b \lvert f\rvert.
   Linearity and monotonicity of the integral (7.1.5a,c, T07) and P05.

4. Since \lvert f(x)\rvert \le M for all x \in [a,b], Theorem 7.1.5(c) and Example 7.1.4(a) give \int_a^b \lvert f\rvert \le \int_a^b M = M(b - a). ∎
   Constant integral (T05).

Converse is false: Define f : [0,1] \to \mathbb{R} by f(x) := 1 if x \in \mathbb{Q}\cap[0,1] and f(x) := -1 if x \in [0,1]\setminus\mathbb{Q}. Then \lvert f(x)\rvert = 1 for all x \in [0,1], so \lvert f\rvert \in

\mathcal{R}[0,1], but f \notin \mathcal{R}[0,1] (rational tags give □S = 1, irrational tags give S = -1).

Marks: \lvert f\rvert \in \mathcal{R}[a,b] via 7.1.6 & 7.3.14 2 · \lvert\int f\rvert \le \int\lvert f\rvert \le M(b-a) 2.5 · counterexample for converse 1.5.

34 PYQ MTS-26 Q6 also MAT-22 Q17, MAT-21 Q18

State and prove the Product Theorem (7.3.16); show \int_a^b fg \ne \left(\int_a^b f\right)\left(\int_a^b g\right) in general.

KNOW FIRST

T30 T28 T08 T07

THEOREM 7.3.16 · PRODUCT THEOREM (BARTLE P. 222)

If f and g belong to \mathcal{R}[a,b], then the product fg belongs to \mathcal{R}[a,b].

PROOF OF THEOREM 7.3.16 (BARTLE P. 222)

1. Since f, g \in \mathcal{R}[a,b], Theorem 7.1.5(b) implies f + g \in \mathcal{R}[a,b]. By the Boundedness Theorem 7.1.6, f, g, and f+g are bounded on [a,b]; choose M > 0 so that f([a,b]), g([a,b]), and (f+g)([a,b]) are all contained in [-M,M].
   Theorem 7.1.5(b) and Boundedness Theorem 7.1.6 (T08).

2. The squaring function \varphi(t) := t^2 is continuous on [-M,M]. By the Composition Theorem 7.3.14, f^2 = \varphi \circ f, g^2 = \varphi \circ g, and (f+g)^2 = \varphi \circ (f+g) all belong to \mathcal{R}[a,b].
   Composition Theorem 7.3.14 (T28) with \varphi(t) = t^2.

3. Since we have the algebraic identity fg = \frac12\big[(f + g)^2 - f^2 - g^2\big], Theorem 7.1.5(a) and (b) imply that fg \in \mathcal{R}[a,b]. ∎
   Linearity of the integral (Theorem 7.1.5, T07).

Why \int_a^b fg \ne \left(\int_a^b f\right)\left(\int_a^b g\right) in general: Let f(x) = g(x) = x on [0,1]. Then \int_0^1 fg = \int_0^1 x^2\,dx = \tfrac13, whereas \left(\int_0^1 f\right)\left(\int_0^1 g\right) = \tfrac12 \cdot \tfrac12 = \tfrac14 \ne \tfrac13.

Marks: statement 1 · f^2, g^2, (f+g)^2 \in \mathcal{R}[a,b] via 7.3.14 2.5 · polarisation identity & 7.1.5 1.5 · counterexample for integral of product 1.

35

State and prove Integration by Parts (Theorem 7.3.17); evaluate \int_0^1 x e^x\,dx.

KNOW FIRST

T31 T30 T20 P11

THEOREM 7.3.17 · INTEGRATION BY PARTS (BARTLE P. 222, EQ. (7))

Let F, G be differentiable on [a,b] and let f := F' and g := G' belong to \mathcal{R}[a,b]. Then \int_a^b fG = \Big[FG\Big]_a^b - \int_a^b Fg.

PROOF OF THEOREM 7.3.17 (BARTLE P. 223)

1. By the Product Rule 6.1.3(c) (P11), the product FG is differentiable on [a,b] and (FG)'(x) = F'(x)G(x) + F(x)G'(x) = f(x)G(x) + F(x)g(x) \quad \text{for all } x \in [a,b].
   Product Rule 6.1.3(c) (P11).

2. Since F and G are differentiable on [a,b], they are continuous on [a,b] (Theorem 6.1.2) and hence belong to \mathcal{R}[a,b] (Theorem 7.2.7). Since f, g \in \mathcal{R}[a,b] by hypothesis, the Product Theorem 7.3.16 implies that fG \in \mathcal{R}[a,b] and Fg \in \mathcal{R}[a,b], so (FG)' = fG + Fg \in \mathcal{R}[a,b] by Theorem 7.1.5(b).
   Theorems 6.1.2, 7.2.7, 7.3.16 (T30), and 7.1.5(b).

3. Applying the First Form of the Fundamental Theorem 7.3.1 (with E = \varnothing) to FG gives \Big[FG\Big]_a^b = \int_a^b (FG)' = \int_a^b fG + \int_a^b Fg, \quad \text{whence} \quad \int_a^b fG = \Big[FG\Big]_a^b - \int_a^b Fg. \quad \blacksquare
   Theorem 7.3.1 (T20) and Theorem 7.1.5(b).

Evaluation of \int_0^1 x e^x\,dx: Put F(x) := e^x (f(x) = F'(x) = e^x \in \mathcal{R}[0,1]) and G(x) := x (g(x) = G'(x) = 1 \in \mathcal{R}[0,1]). By Theorem 7.3.17, \int_0^1 e^x\cdot x\,dx = \big[x e^x\big]_0^1 - \int_0^1 e^x\cdot 1\,dx = (1\cdot e - 0) \big[e^x\big]_0^1 = e - (e - 1) = 1.

ANSWER $\displaystyle\int_0^1 x e^x\,dx = 1.$

Marks: statement 1 · Product Rule + Product Theorem step 2 · FTC 1 conclusion 1.5 · \int_0^1 x e^x\,dx = 1 1.5.

36 PYQ MTS-25 Q19 also MAT-22 Q32

State and prove Taylor's Theorem with the Integral Remainder (7.3.18); verify for f(x) = e^x on [0,1] with n = 1, 2.

KNOW FIRST

T32 T31 T20

THEOREM 7.3.18 · TAYLOR'S THEOREM WITH THE REMAINDER (BARTLE P. 223, EQS. (8)–(9))

Suppose that f', \dots, f^{(n)}, f^{(n+1)} exist on [a,b] and that f^{(n+1)} \in \mathcal{R}[a,b]. Then we have f(b) = f(a) + \frac{f'(a)}{1!}(b - a) + \cdots + \frac{f^{(n)}(a)}{n!}(b - a)^n + R_n, where the remainder is given by R_n = \frac{1}{n!}\int_a^b f^{(n+1)}(t)\,(b - t)^n\,dt.

PROOF OF THEOREM 7.3.18 (BARTLE P. 223, WITH MISPRINT CORRECTED)

1. Apply Integration by Parts (Theorem 7.3.17) to R_n with F(t) := f^{(n)}(t) (so f(t) = F'(t) = f^{(n+1)}(t) \in \mathcal{R}[a,b]) and G(t) := (b - t)^n/n! (so g(t) = G'(t) = -(b - t)^{n-1}/(n - 1)! \in \mathcal{R}[a,b]).
   Check hypotheses of Theorem 7.3.17 (T31).

2. Since G(b) = 0 and G(a) = (b - a)^n/n!, Theorem 7.3.17 gives R_n = \left[\frac{f^{(n)}(t)(b-t)^n}{n!}\right]_{t=a}^{t=b} - \int_a^b f^{(n)}(t)\left(-\frac{(b-t)^{n-1}}{(n-1)!}\right)dt = -\frac{f^{(n)}(a)}{n!}(b - a)^n + R_{n-1}, where R_{n-1} = \dfrac{1}{(n-1)!}\int_a^b f^{(n)}(t)\,(b - t)^{n-1}\,dt.
   Boundary evaluation at t=b and t=a. (Note: Bartle p. 223 prints (b-a)^{n-1} in the first displayed integral; the correct integrand is (b-t)^{n-1}.)

3. For each k \le n, f^{(k)} is differentiable on [a,b], hence continuous (Theorem 6.1.2) and in \mathcal{R}[a,b] (Theorem 7.2.7). Repeating Step 2 n times yields R_{k-1} = \frac{f^{(k)}(a)}{k!}(b-a)^k + R_k down to R_0 = \int_a^b f'(t)\,dt = f(b) - f(a) (by FTC 1, Theorem 7.3.1), which rearranges to Taylor's formula. ∎
   Repeated Integration by Parts and FTC 1 (T20).

Verification for f(x) = e^x, a = 0, b = 1: Here f^{(k)}(0) = 1 for all k. For n = 1: T_1 = 1 + 1 = 2 and R_1 =

\int_0^1 e^t(1-t)\,dt = \big[e^t(1-t)\big]_0^1 + \int_0^1 e^t\,dt = -1 + (e - 1) = e - 2 \approx 0.718282, so T_1 + R_1 = e. For n = 2: T_2 = 1 + 1 + \tfrac12 = \tfrac52 and R_2 = -\tfrac12 + R_1 = e - \tfrac52 \approx 0.218282, so T_2 + R_2 = e.

ANSWER **R_1 = e - 2 \approx 0.718282;\quad R_2 = e - \frac52 \approx 0.218282.**

Marks: statement 1.5 · IBP recurrence R_n = -\frac{f^{(n)}(a)}{n!}(b-a)^n + R_{n-1} 2 · induction to R_0 = f(b)-f(a) 1 · e^x check 1.5.

Section C · Essay answers (Q37–Q40)

10 marks each. Answer any two.

37 PYQ MTS-25 Q18 also MAT-21 Q30, MTS-22 Q19, MAT-22 MCQ 21, MAT-21 MCQ 15

Continuous on [a,b] \Rightarrow Riemann integrable □(Theorem 7.2.7); where hypotheses are used; converses.

KNOW FIRST

T15 T12 T14 P07 P08

THEOREM 7.2.7 · INTEGRABILITY OF CONTINUOUS FUNCTIONS (BARTLE P. 212)

If f : [a,b] \to \mathbb{R} is continuous on [a,b], then f \in \mathcal{R}[a,b].

SKELETON

1. Use UCT 5.4.3 to get \delta_\varepsilon > 0 so \lvert u - v\rvert < \delta_\varepsilon \Rightarrow \lvert f(u) - f(v)\rvert < \varepsilon/(b-a). 2. Pick a partition P with \lVert P\rVert < \delta_\varepsilon. 3. Use Max–Min 5.3.4 on each I_i to get u_i, v_i \in I_i with f(u_i) = \min_{I_i} f, f(v_i) = \max_{I_i} f. 4. Define step functions \alpha_\varepsilon, \omega_\varepsilon with heights f(u_i), f(v_i) and apply the Squeeze Theorem 7.2.3.

PROOF OF THEOREM 7.2.7 (BARTLE P. 212)

1. Since f is continuous on the closed bounded interval [a,b], the Uniform Continuity Theorem 5.4.3 (P08) implies that f is uniformly continuous on [a,b]. Therefore, given any \varepsilon > 0, there exists \delta_\varepsilon > 0 such that if u, v \in [a,b] and \lvert u - v\rvert < \delta_\varepsilon, then \lvert f(u) - f(v)\rvert < \frac{\varepsilon}{b - a}.
   Uniform Continuity Theorem 5.4.3 applied with tolerance \varepsilon/(b-a) > 0.

2. Let P = \{I_i\}_{i=1}^n be a partition of [a,b] with \lVert P\rVert < \delta_\varepsilon, where I_i = [x_{i-1},x_i].
   Choose an equal partition with n > (b-a)/\delta_\varepsilon.

3. Since f is continuous on each closed bounded subinterval I_i = [x_{i-1},x_i], the Maximum–Minimum Theorem 5.3.4 (P07) implies that there exist points u_i, v_i \in I_i where f attains its minimum and maximum values on I_i, respectively: f(u_i) \le f(x) \le f(v_i) \quad \text{for all } x \in I_i.
   Maximum–Minimum Theorem 5.3.4 on each I_i.

4. Define step functions \alpha_\varepsilon, \omega_\varepsilon : [a,b] \to \mathbb{R} by \alpha_\varepsilon(x) := f(u_i) \text{ for } x \in [x_{i-1},x_i)\ (i < n), \quad \alpha_\varepsilon(x) := f(u_n) \text{ for } x \in [x_{n-1},x_n], \omega_\varepsilon(x) := f(v_i) \text{ for } x \in [x_{i-1},x_i)\ (i < n), \quad \omega_\varepsilon(x) := f(v_n) \text{ for } x \in [x_{n-1},x_n]. By Theorem 7.2.5, \alpha_\varepsilon, \omega_\varepsilon \in \mathcal{R}[a,b], and by Step 3, \alpha_\varepsilon(x) \le f(x) \le \omega_\varepsilon(x) for all x \in [a,b].
   Step-function integrability (Theorem 7.2.5, T14).

5. Since u_i, v_i \in I_i and \lVert P\rVert < \delta_\varepsilon, we have \lvert u_i v_i\rvert \le x_i - x_{i-1} < \delta_\varepsilon, so f(v_i) - f(u_i) < \dfrac{\varepsilon}{b-a} by Step 1. Therefore, by formula (5) of Theorem 7.2.5, 0 \le \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) = \sum_{i=1}^n \big(f(v_i) - f(u_i)\big)(x_i - x_{i-1}) < \sum_{i=1}^n \frac{\varepsilon}{b - a}(x_i - x_{i-1}) = \frac{\varepsilon}{b - a}(b - a) = \varepsilon.
   Theorem 7.2.5 formula and telescoping \sum(x_i - x_{i-1}) = b - a (M01).

6 Since \varepsilon > 0 is arbitrary, the Squeeze Theorem 7.2.3 implies that f \in \mathcal{R}[a,b]. ∎

Squeeze Theorem 7.2.3 (T12).

Where each hypothesis was used: Continuity of f and the fact that [a,b] is closed and bounded are used twice: (i) in Step 1 to invoke the Uniform Continuity Theorem 5.4.3 so a single \delta_\varepsilon works across all subintervals, and (ii) in Step 3 to invoke the Maximum–Minimum Theorem 5.3.4 on each closed subinterval I_i.

Converses are FALSE (MAT-22 MCQ 21, MAT-21 MCQ 15): Option (A) "f continuous on [a,b] \Rightarrow f \in \mathcal{R}[a,b]" is true, while (B) and (C) are false: the step function g(x) = 2 on [0,1], 3 on (1,3] (or Thomae's function h C8) belongs to \mathcal{R}, yet is discontinuous and hence not uniformly continuous.

Marks: UCT step 2 · Max–Min step 2 · step functions \alpha_\varepsilon, \omega_\varepsilon 2 · \int(\omega_\varepsilon-\alpha_\varepsilon) < \varepsilon & Squeeze 2 · where hypotheses used & converses 2.

38 PYQ MTS-26 Q19 also MAT-22 Q30, MAT-22 MCQ 22

Monotone on [a,b] \Rightarrow Riemann integrable (Theorem 7.2.8); staircase calculation for f(x) = x^2 on [0,1] (n=4); converse.

KNOW FIRST

T16 T12 T14 M05

THEOREM 7.2.8 · INTEGRABILITY OF MONOTONE FUNCTIONS (BARTLE P. 212)

If f : [a,b] \to \mathbb{R} is monotone on [a,b], then f \in \mathcal{R}[a,b].

SKELETON

Assume f is increasing on [a,b]. Partition [a,b] into n equal subintervals of length (b-a)/n. Define step functions \alpha(x) = f(x_{k-1}) and \omega(x) = f(x_k) on I_k, telescope \int_a^b(\omega-\alpha) = \frac{b-a}{n}(f(b)-f(a)), choose n > (b-a)(f(b)-f(a))/\varepsilon, and apply the Squeeze Theorem 7.2.3.

PROOF OF THEOREM 7.2.8 (BARTLE P. 212)

1. Assume f is increasing on I = [a,b] (if f is decreasing, then -f is increasing, so -f \in \mathcal{R}[a,b] and hence f = -(-f) \in \mathcal{R}[a,b] by Theorem 7.1.5(a)). For n \in \mathbb{N}, partition [a,b] into n equal subintervals I_k = [x_{k-1},x_k] with x_k - x_{k-1} = \dfrac{b-a}{n} (k = 1, \dots, n).
   Equal subinterval lengths (b-a)/n allow the step integrals to telescope.

2. Since f is increasing on I_k = [x_{k-1},x_k], its minimum on I_k is f(x_{k-1}) and its maximum on I_k is f(x_k): f(x_{k-1}) \le f(x) \le f(x_k) \quad \text{for all } x \in I_k.
   Definition of an increasing function (P10).

3. Define step functions \alpha, \omega : [a,b] \to \mathbb{R} by \alpha(x) := f(x_{k-1}) and \omega(x) := f(x_k) for x \in [x_{k-1},x_k) (k = 1,\dots,n-1), and \alpha(x) := f(x_{n-1}), \omega(x) := f(x_n) for x \in [x_{n-1},x_n]. Then \alpha, \omega \in \mathcal{R}[a,b] by Theorem 7.2.5 and \alpha(x) \le f(x) \le \omega(x) for all x \in [a,b].
   Theorem 7.2.5 (T14).

4. By Theorem 7.2.5, $\int_a^b \alpha = \frac{b - a}{n}\big(f(x_0) + f(x_1) + \cdots + f(x_{n-1})\big)$, $\qquad \int_a^b \omega = \frac{b - a}{n}\big(f(x_1) + \cdots + f(x_{n-1}) + f(x_n)\big)$. Subtracting and cancelling the interior terms $f(x_1), \dots, f(x_{n-1})$ yields $\int_a^b (\omega - \alpha) = \frac{b - a}{n}\big(f(x_n) - f(x_0)\big) = \frac{b - a}{n}\big(f(b) f(a)\big)$.
   Telescoping sum (M01, M05) with $x_0 = a$ and $x_n = b$.

5. Given $\varepsilon > 0$, choose $n \in \mathbb{N}$ with $n > \dfrac{(b - a)(f(b) f(a))}{\varepsilon}$. Then $\int_a^b (\omega - \alpha) < \varepsilon$, so by the Squeeze Theorem 7.2.3, $f \in \mathcal{R}[a,b]$. ∎
   Archimedean property (P13) and Squeeze Theorem 7.2.3 (T12).

Staircase illustration for $f(x) = x^2$ on $[0,1]$ with $n = 4$: Here $P_4 = (0, \tfrac14, \tfrac12, \tfrac34, 1)$ has width $\Delta x_k = \tfrac14$, and $f(x) = x^2$ takes values $0, \tfrac1{16}, \tfrac4{16}, \tfrac9{16}, 1$ at the partition points:

$$\int_0^1 \alpha = \frac14\left(0 + \frac1{16} + \frac4{16} + \frac9{16}\right) = \frac{14}{64} = \frac{7}{32} = 0.21875, \qquad \int_0^1 \omega = \frac14\left(\frac1{16} + \frac4{16} + \frac9{16} + 1\right) = \frac{30}{64} = \frac{15}{32} = 0.46875,$$

and $\int_0^1(\omega - \alpha) = 0.46875 - 0.21875 = 0.25 = \frac{1-0}{4}(1 - 0)$.

Converse is FALSE (MAT-22 MCQ 22): Option (A) is true, while (B) "f \in \mathcal{R}[a,b] \Rightarrow f is monotone" is false: f(x) = \sin x on [0,\pi] (or x^2 on [-1,1], or Thomae's function C8) is in \mathcal{R} but not monotone.

ANSWER **For x^2 on [0,1] (n=4): \int_0^1 \alpha = 0.21875, \int_0^1 \omega = 0.46875, difference = 0.25.**

Marks: equal partition & \alpha,\omega setup 2.5 · telescoping \int(\omega-\alpha) = \frac{b-a}{n}(f(b)-f(a)) 2.5 · choice of n & Squeeze 2 · x^2 (n=4) calculation 2 · converse false 1.

39 PYQ MTS-26 Q25 also MAT-22 Q37a, MTS-22 Q25

State and prove the Additivity Theorem (Theorem 7.2.9).

KNOW FIRST

T17 T10 T08 T03 M08

THEOREM 7.2.9 · ADDITIVITY THEOREM (BARTLE P. 213)

Let f : [a,b] \to \mathbb{R} and let c \in (a,b). Then f \in \mathcal{R}[a,b] if and only if its restrictions to [a,c] and to [c,b] are both Riemann integrable. In this case \int_a^b f = \int_a^c f + \int_c^b f.

SKELETON

(\Leftarrow) Take \delta', \delta'' for f_1, f_2 with \varepsilon/3; state f is bounded by M > 0 (7.1.6 on each piece); put \delta_\varepsilon = \min\{\delta',\delta'',\varepsilon/(6M)\}. Split into Case (i) c \in Q and Case (ii) c \in (x_{k-1},x_k). (\Rightarrow) Use Cauchy 7.2.1 by extending \dot P_1, \dot Q_1 on [a,c] with the same \dot R on [c,b].

PROOF OF THEOREM 7.2.9 (BARTLE PP. 213–214)

(\Leftarrow)

1. Suppose f_1 := f|_{[a,c]} \in \mathcal{R}[a,c] with integral L_1 and f_2 := f|_{[c,b]} \in \mathcal{R}[c,b] with integral L_2. Given \varepsilon > 0, choose \delta' > 0 such that \lVert P_1\rVert < \delta' \Rightarrow \lvert S(f_1;\dot P_1) - L_1\rvert < \varepsilon/3, and choose \delta'' > 0 such that \lVert P_2\rVert < \delta'' \Rightarrow \lvert S(f_2;\dot P_2) - L_2\rvert < \varepsilon/3.
   Definition 7.1.1 applied on [a,c] and [c,b] with tolerance \varepsilon/3.

2. Since f_1 \in \mathcal{R}[a,c] and f_2 \in \mathcal{R}[c,b], Theorem 7.1.6 implies that f is bounded on [a,c] and on [c,b], and hence bounded on [a,b]. Choose M > 0 such that \lvert f(x)\rvert \le M for all x \in [a,b] (if f \equiv 0, take any M > 0), and define \delta_\varepsilon := \min\left\{\delta',\, \delta'',\, \frac{\varepsilon}{6M}\right\} > 0. Let \dot Q = \{(I_i,t_i)\}_{i=1}^m be any tagged partition of [a,b] with \lVert Q\rVert < \delta_\varepsilon.
   Boundedness Theorem 7.1.6 (T08) and three-way minimum (M08).

3 **Case (i): c is a partition point of \dot Q.** Split \dot Q at c into a tagged partition \dot Q_1 of [a,c] and a tagged partition \dot Q_2 of [c,b]. Since S(f;\dot Q) = S(f_1;\dot Q_1) + S(f_2;\dot Q_2) and \lVert Q_1\rVert < \delta', \lVert Q_2\rVert < \delta'', Step 1 and the Triangle Inequality give \lvert S(f;\dot Q) - (L_1 + L_2)\rvert \le \lvert S(f_1;\dot Q_1) - L_1\rvert + \lvert S(f_2;\dot Q_2) L_2\rvert < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} < \varepsilon.

Direct split when c is a cut point.

4 **Case (ii): c is not a partition point of \dot Q.** Then c \in (x_{k-1},x_k) for some k \le m. Define tagged partitions \dot Q_1 of [a,c] and \dot Q_2 of [c,b] by \dot Q_1 := \{(I_1,t_1),\dots,(I_{k-1},t_{k-1}),([x_{k-1},c],c)\}, \qquad \dot Q_2 := □\{([c,x_k],c), (I_{k+1},t_{k+1}),\dots,(I_m,t_m)\}. Then all terms of S(f;\dot Q) and S(f_1;\dot Q_1) + S(f_2;\dot Q_2) cancel except on I_k, giving S(f;\dot Q) - S(f_1;\dot Q_1) - S(f_2;\dot Q_2) = \big(f(t_k) - f(c)\big)(x_k - x_{k-1}), so \lvert S(f;\dot Q) - S(f_1;\dot Q_1) - S(f_2;\dot Q_2)\rvert \le 2M(x_k - x_{k-1}) \le 2M\lVert Q\rVert < 2M\cdot\dfrac{\varepsilon}{6M} = \dfrac{\varepsilon}{3}.

Only interval I_k is modified; \lvert f(t_k) - f(c)\rvert \le 2M and x_k - x_{k-1} < \varepsilon/(6M).

5. Since \lVert Q_1\rVert \le \lVert Q\rVert < \delta' and \lVert Q_2\rVert \le \lVert Q\rVert < \delta'', Step 1 gives \lvert S(f_1;\dot Q_1) L_1\rvert < \varepsilon/3 and \lvert S(f_2;\dot Q_2) - L_2\rvert < \varepsilon/3. By the Triangle Inequality, \lvert S(f;\dot Q) - (L_1 + L_2)\rvert \le \lvert S(f;\dot Q) - S(f_1;\dot Q_1) - S(f_2;\dot Q_2)\rvert + \lvert S(f_1;\dot Q_1) - L_1\rvert + \lvert S(f_2;\dot Q_2) - L_2\rvert < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. Since \varepsilon > 0 is arbitrary, f \in \mathcal{R}[a,b] and \int_a^b f = L_1 + L_2 = \int_a^c f + \int_c^b f.
   Triangle Inequality and Definition 7.1.1.

(\Rightarrow)

1. Suppose f \in \mathcal{R}[a,b] and let \varepsilon > 0. By the Cauchy Criterion 7.2.1, there exists \eta_\varepsilon > 0 such that any tagged partitions \dot P, \dot Q of [a,b] with norms <\eta_\varepsilon satisfy \lvert S(f;\dot P) - S(f;\dot Q)\rvert < \varepsilon.
   Cauchy Criterion 7.2.1 (\Rightarrow).

2 Let \dot P_1, \dot Q_1 be any tagged partitions of [a,c] with \lVert P_1\rVert < \eta_\varepsilon and \lVert Q_1\rVert < \eta_\varepsilon. Adjoin the **same** tagged partition \dot R of [c,b] with \lVert R\rVert < \eta_\varepsilon to both \dot P_1 and \dot Q_1 to obtain tagged partitions \dot P = \dot P_1 \cup \dot R and \dot Q = \dot Q_1 \cup \dot R of [a,b] with \lVert P\rVert, \lVert Q\rVert < \eta_\varepsilon. Then \lvert S(f_1;\dot P_1) - S(f_1;\dot Q_1)\rvert = \lvert S(f;\dot P) - S(f;\dot Q)\rvert < \varepsilon. By the Cauchy Criterion 7.2.1, f_1 \in \mathcal{R}[a,c]. By the same argument, f_2 \in \mathcal{R}[c,b], and \int_a^b f = \int_a^c f + \int_c^b f follows from the (\Leftarrow) part. ∎

The [c,b] terms S(f;\dot R) cancel identically; Cauchy Criterion 7.2.1 (\Leftarrow).

Marks: statement 1 · boundedness M>0 & \delta_\varepsilon = \min\{\delta',\delta'',\varepsilon/(6M)\} 2 · Case (i) & Case (ii) estimates 4 · (\Rightarrow) Cauchy extension 3.

40 PYQ MTS-25 Q25 also MAT-21 Q37a, MAT-22 Q33

State and prove the Second Form of the Fundamental Theorem of Calculus (Theorems 7.3.5 & 7.3.6); counterexample at a discontinuity.

KNOW FIRST

T23 T24 T22 T17 T07

THEOREM 7.3.5 & THEOREM 7.3.6 · FTC SECOND FORM (BARTLE P. 219)

Theorem 7.3.5: Let f \in \mathcal{R}[a,b] and let f be continuous at a point c \in [a,b]. Then the indefinite integral F(z) := \int_a^z f (z \in [a,b]) is differentiable at c and F'(c) = f(c).

Theorem 7.3.6: If f is continuous on [a,b], then the indefinite integral F(z) := \int_a^z f is differentiable on [a,b] and F'(x) = f(x) for all x \in [a,b].

SKELETON

For c \in [a,b), given \varepsilon > 0 use continuity of f at c to find \eta_\varepsilon > 0 so c \le x < c + \eta_\varepsilon \Rightarrow f(c) - \varepsilon < f(x) < f(c) + \varepsilon. For 0 < h < \eta_\varepsilon, write F(c+h) - F(c) = \int_c^{c+h} f, sandwich between (f(c)-\varepsilon)h and (f(c)+\varepsilon)h, divide by h, and repeat for h < 0.

PROOF OF THEOREMS 7.3.5 AND 7.3.6 (BARTLE PP. 219–220)

1. Suppose first that $c \in [a,b)$ and consider the right-hand derivative of F at c. Since f is continuous at c, given $\varepsilon > 0$ there exists $\eta_\varepsilon > 0$ such that if $c \le x < c + \eta_\varepsilon$ ($x \in [a,b]$), then $f(c) - \varepsilon < f(x) < f(c) + \varepsilon$.
   Definition 5.1.1 of continuity of f at c.

2. Let h satisfy $0 < h < \eta_\varepsilon$ (with $c + h \le b$). By the Additivity Theorem 7.2.9 (and Corollary 7.2.10), f is integrable on $[a,c]$, $[a,c+h]$, and $[c,c+h]$, and $F(c + h) - F(c) = \int_a^{c+h} f - \int_a^c f = \int_c^{c+h} f$.
   Additivity Theorem 7.2.9 (T17).

3. On the interval $[c, c+h]$, the inequality of Step 1 holds, so Theorem 7.1.5(c) and Example 7.1.4(a) yield $(f(c) - \varepsilon)\cdot h \le F(c + h) - F(c) = \int_c^{c+h} f \le (f(c) + \varepsilon)\cdot h$.
   Monotonicity of the integral (Theorem 7.1.5(c), T07) and constant integral $\int_c^{c+h} k = kh$ (T05).

4. Dividing this inequality by $h > 0$ and subtracting $f(c)$ gives $\left\lvert\frac{F(c + h) - F(c)}{h} - f(c)\right\rvert \le \varepsilon$. Since $\varepsilon > 0$ is arbitrary, the right-hand limit is $\lim_{h\to 0^+} \dfrac{F(c+h) - F(c)}{h} = f(c)$.
   Definition of right-hand derivative.

5. If $c \in (a,b]$, the left-hand limit is proved identically: for $-\eta_\varepsilon < h < 0$ (where $f(c)-\varepsilon < f(x) < f(c)+\varepsilon$ on $(c-\eta_\varepsilon,c]$), on $[c+h,c]$ of length $-h > 0$ we have $(f(c)-\varepsilon)(-h) \le \int_{c+h}^c f = F(c) - F(c+h) \le (f(c)+\varepsilon)(-h)$; dividing by $-h > 0$ gives $\left\lvert\frac{F(c+h)-F(c)}{h} - f(c)\right\rvert \le \varepsilon$, so $\lim_{h\to 0^-} \frac{F(c+h)-F(c)}{h} = f(c)$. Hence F is differentiable at c and $F'(c) = f(c)$.
   Agreement of left and right derivatives at interior points (and one-sided derivative at a or b).

6. **Proof of Theorem 7.3.6:** If f is continuous at every point of $[a,b]$, then $f \in \mathcal{R}[a,b]$ by Theorem 7.2.7, and applying Theorem 7.3.5 at each $x \in [a,b]$ shows that F is differentiable on $[a,b]$ with $F'(x) = f(x)$ for all $x \in [a,b]$. ∎
   Pointwise application of Theorem 7.3.5 across $[a,b]$. (Note: MAT-22 Q33 states Theorem 7.3.5 in full.)

Example where F fails to be differentiable at 0 (Example 7.3.7(a), C20): Let f(x) := \operatorname{sgn} x on [-1,1]. Then f \in \mathcal{R}[-1,1] (step function), and its indefinite integral with basepoint -1 is F(x) = \int_{-1}^x \operatorname{sgn} = \lvert x\rvert - 1. At c = 0 (where f is discontinuous), F'_-(0) = -1 \ne +1 = F'_+(0), so F is not differentiable at 0.

Marks: statements of 7.3.5 & 7.3.6 2 · continuity of f at c setup 1.5 · Additivity F(c+h)-F(c) = \int_c^{c+h} f 1.5 · sandwiching & division by h 2 · left-hand limit & 7.3.6 1.5 · \operatorname{sgn} counterexample 1.5.